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Explainable Edge-AI Framework for Fault Diagnosis and Remaining Useful Life Prediction in Hydraulic Systems

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Abstract

Predictive maintenance solutions for hydraulic systems often fail to reach industrial adoption, not due to insufficient model accuracy, but because of fragmented data infrastructures, missing operational context, and limited physical interpretability. This work proposes an edge-oriented, open-source framework for condition monitoring and Remaining Useful Life (RUL) estimation that addresses the minimum

architectural and data requirements for production-grade deployment. The framework is designed as an event-driven and state-based pipeline, where operational events, component behavior, and sensor data are consistently aligned.

Using a real hydraulic system controlled by an industrial PLC, machine states are identified through unsupervised clustering and Hidden Markov Models, enabling degradation-aware state transitions without explicit fault labels. RUL is formulated as the time to transition into economically non-viable operational states rather than catastrophic failure. Prognostic models based on LSTM with attention and XGBoost are complemented by explainability analyses (SHAP, PDP, attention maps) that link model outputs to physically meaningful hydraulic variables, enabling expert validation and operational trust.

Experimental results show that the identified states and predicted RUL trajectories are consistent with logged downtime events and known physical degradation mechanisms, while explainability analyses confirm that statistically relevant features correspond to physically meaningful variables.

The proposed architecture defines the minimum architectural structure required for edge-oriented deployment, enabling scalable, interpretable, and autonomous predictive maintenance solutions with direct relevance for production availability and Overall Equipment Effectiveness (OEE).

Keywords: Predictive Maintenance, Intelligent Diagnosis, hydraulic systems, Remaining Useful Life (RUL), Long Short-Term Memory (LSTM), Hidden Markov Models (HMM).

Table 1: List of nomenclature

Nomenclature	Description
AI	Artificial Intelligence
ACR	Asset Criticality Ranking
CPS	Cyber-Physical System
ctrlX CORE	Programmable Logic Controller from Bosch Rexroth
DL	Deep Learning
Docker	Platform for developing, shipping, and running containerized applications
Dagster	Data orchestration platform for building and managing data pipelines
ERP	Enterprise Resource Planning
ETL	Extract, Transform, Load
FMECA	Failure Mode, Effects, and Criticality Analysis
GUI	Graphical User Interface
HMM	Hidden Markov Model
Industry 4.0	Fourth Industrial Revolution
LSTM	Long Short-Term Memory
MES	Manufacturing Execution System
ML	Machine Learning
OPC UA	Open Platform Communications Unified Architecture
PdM	Predictive Maintenance
PDP	Partial Dependence Plot (global marginal effect visualization method)
PLC	Programmable Logic Controller
PHM	Prognostics and Health Management
PostgreSQL	Object-relational database management system
Power BI	Business Intelligence tool from Microsoft
ROI	Return On Investment
RUL	Remaining Useful Life
SAP	Systems, Applications, and Products in Data Processing
SAP PM	SAP Plant Maintenance
SHAP	SHapley Additive exPlanations
Solace	Middleware platform for event-driven streaming and message brokering
XGBoost	Extreme Gradient Boosting algorithm

1 Introduction

The Fourth Industrial Revolution, as presented by Lasi et al. [1], describes the IT-driven transformation of manufacturing systems, involving both technological and organizational innovation. Within this context, CPS enable new paradigms in production, where PdM plays a pivotal role. PdM relies on continuous monitoring of system conditions through sensor networks, which capture real-time operational data for subsequent analysis.

According to Zonta et al. [2], PdM harnesses historical records, domain knowledge, and computational models to forecast the behavior of critical assets. By applying statistical and machine learning methods, it identifies trends and anomalies that may indicate early stages of degradation or failure. This predictive capability facilitates proactive decision-making and helps reduce unplanned downtime.

As highlighted in [3], establishing an ACR is essential for identifying components that justify further FMECA. Once identified, these critical assets are subject to more advanced diagnostic and predictive strategies, optimizing the allocation of maintenance resources.

Dalzochio et al. [4] further emphasize that the deployment of ML-based PdM solutions in industrial environments faces core challenges related to data volume, latency, scalability, and bandwidth constraints. While edge computing is often proposed

as a mitigation strategy, PdM does not inherently require real-time model execution. Instead, periodic analysis—based on aggregated data collected over defined intervals—is often sufficient to detect early signs of degradation [5].

As Pérez et al. [6] point out, critical events such as sudden failures or fast deviations from setpoints are typically addressed by built-in threshold-based alarms at the PLC level, which operate on millisecond timescales. These mechanisms provide immediate warnings to ensure process integrity and safety. In contrast, components undergoing slow degradation—often undetectable by standard alarms—are more effectively assessed using ML and DL models trained on longer time horizons. This functional separation supports a two-tiered monitoring approach: real-time visibility for safety and periodic analytics for predictive insights.

Contribution of this work. This paper introduces an edge-oriented, data-driven framework for predictive maintenance in hydraulic systems, addressing challenges related to data variability, multirate sampling, and scalable industrial deployment. Rather than focusing on isolated algorithmic advances, the proposed approach emphasizes the systematic integration of machine learning, signal processing, and industrial data engineering within a deployable and interpretable edge architecture, thereby bridging the gap between research-oriented models and production-ready solutions.

The contributions are threefold. (1) We define a hybrid modeling strategy that combines unsupervised clustering (K-Means) and probabilistic sequence modeling (Hidden Markov Models) for data-driven machine state identification, providing a structured basis for degradation-aware monitoring without requiring explicit fault labels. (2) We formulate RUL estimation as a state-transition forecasting problem, implemented using an LSTM network with an attention mechanism and benchmarked against an XGBoost regressor, enabling the modeling of temporal degradation dynamics from multivariate hydraulic time-series data. (3) We design and experimentally validate a fully deployable edge-oriented framework on a real hydraulic system controlled by a Bosch Rexroth `ctrlX CORE` PLC with embedded OPC UA communication, demonstrating industrial feasibility, interpretability, and operational autonomy.

Experimental results indicate that the proposed framework enables continuous and robust RUL estimation under varying operating conditions, while explainability analyses link model outputs to physically meaningful hydraulic variables. The findings highlight how a state-based prognostic formulation combined with edge-oriented system design can support scalable and trustworthy PHM solutions in complex manufacturing environments.

In this work, *edge-oriented* refers to the deployment of data acquisition and analytics services in close proximity to the PLC, ensuring operational autonomy. Cloud connectivity, when present, is optional and used only for scalability or offline analysis.

Paper organization. The remainder of this paper is structured as follows. Section 2 reviews related work on data-driven predictive maintenance, hydraulic fault diagnosis, and edge-based industrial architectures. Section 3 describes the investigated hydraulic system, including its physical configuration, failure modes, and the PLC-oriented modeling abstraction. Section 4 presents the proposed edge-oriented data architecture, covering event logging, database design, and the ETL pipeline for multi-rate sensor

acquisition. Section 5 details the machine learning and deep learning methods used for state classification and RUL estimation, including feature engineering, model training, and explainability. Section 7 discusses results and implications, and Section 8 concludes the paper.

2 Related Work

The escalating costs associated with unplanned downtime underscore the need for proactive maintenance strategies that can both (i) detect incipient degradation and (ii) support actionable decision-making in realistic industrial settings. In this context, Prognostics and Health Management (PHM) aims to maximize RUL while mitigating unforeseen failures. However, practical deployment remains constrained by data availability, heterogeneous acquisition layers, multirate sensing, and the lack of architectural guidance for scalable and interoperable integration across industrial IT/OT boundaries [7].

2.1 Electro-hydraulic modeling and control foundations

Electro-hydraulic actuators are strongly nonlinear systems exhibiting coupled pressure–flow–motion dynamics, motivating physics-based modeling as a reference for control, monitoring, and diagnostic abstraction. Representative studies derive continuous-time formulations using force balance, chamber continuity, and orifice-based valve flow relations, and validate them under realistic operating conditions. Sahu *et al.* characterize the static and dynamic behavior of a high-performance electro-hydraulic actuator and assess closed-loop performance for industrial suitability [8]. Xin *et al.* establish valve-controlled mathematical models for differential cylinders, deriving transfer functions under elastic and non-elastic loads, thereby providing a theoretical basis for analyzing static and dynamic characteristics [9]. From an implementation-oriented perspective, Adnan *et al.* demonstrate system identification from open-loop experimental data and real-time controller design via pole placement, highlighting the practicality of data-driven identification when full physical parametrization is infeasible [10]. Robustness considerations are emphasized by Niksefat and Sepehri, who design a force controller resilient to structured plant and environmental uncertainties using Quantitative Feedback Theory (QFT), illustrating the importance of uncertainty-aware designs in real deployments [11]. Collectively, these works motivate the PLC-oriented abstraction adopted in this paper, where discrete valve commands excite an underlying continuous plant and measured signals are mapped consistently to model variables for monitoring and learning.

Related integrated formulations can also be found in electro-hydraulic actuator (EHA) concepts, where motor, pump, and actuator dynamics are jointly represented within a unified hydrostatic model, supporting compact abstractions of coupled pressure–flow–motion behavior for monitoring purposes [12]. Recent simulation-based investigations further analyze circuit architectures and controller strategies for electro-hydrostatic actuators in mobile machinery, highlighting system-level performance and control-oriented design aspects [13, 14].

2.2 Data acquisition, synchronization, and industrial interoperability

While algorithmic advances in PHM are substantial, many studies under-specify how analysis-ready industrial datasets are acquired, synchronized, and maintained at scale. Nentwich and Reinhart highlight that PdM performance is tightly coupled to acquisition strategy design, proposing a systematic methodology to select monitored components, excitation trajectories, and sampling frequencies [15]. For hydraulic machinery, Qiu *et al.* explicitly address synchronization, anti-interference, and deployment complexity, proposing a fault-diagnosis-oriented acquisition system validated in power stations and pump test platforms [16]. Recent hardware-level contributions further emphasize scalable and synchronized measurement infrastructures: Toscani *et al.* propose a daisy-chain digital bus architecture supporting heterogeneous transducers with guaranteed synchronization for condition monitoring [17]. Beyond signal acquisition, interoperability of analytics models across manufacturing layers remains a critical barrier. Shin proposes an OPC UA-compliant interface for exchanging predictive models (PMML) via standardized information models, enabling publish/subscribe semantics for regression and neural-network models and fostering reusable industrial intelligence [18]. Ontology-driven integration approaches further address semantic consistency in cyber-physical production systems: Nuñez and Borsato introduce an ontology-based PHM model aligned with CPS architectures [19], while Brecher *et al.* demonstrate ontology-based data management for adaptable safety functions in distributed production environments [20]. These works collectively reinforce the architectural focus of the present paper, where OPC UA-based acquisition and edge-oriented ETL are treated as first-class design elements rather than implementation details.

2.3 Data-driven PdM: learning from data correlations and practical limitations

Data-driven predictive maintenance methods are grounded in machine learning, within which deep learning constitutes a specialized subset rather than an independent paradigm. Classical machine learning refers to feature-based models such as linear regression, support vector machines, decision trees, and ensemble methods, which operate on explicitly engineered descriptors.

Comprehensive surveys report a methodological shift within machine learning, from classical feature-driven approaches towards deep learning models capable of automated representation learning as data volumes and system complexity increase [21]. A structured overview of these learning paradigms, model families, and their inductive biases is provided in Table 3.

Despite their expressive capacity, data-driven models fundamentally learn statistical correlations present in the available data. Their predictive performance is therefore inherently constrained by the representativeness of the observed operating conditions and by whether physically meaningful degradation patterns are reflected in the data distribution. When relevant physical relationships are weak, masked by noise, or absent from the measured signals, purely data-driven approaches cannot recover them without additional structural information.

In realistic industrial environments, sensor data are multidimensional, noisy, and often nonstationary. Prior work highlights the importance of robust feature construction through trend extraction, unsupervised selection, and signal decomposition before prognostic modeling [22]. Such considerations underline the central role of preprocessing and feature engineering in data-driven PdM.

For contextualization, Table 2 summarizes commonly adopted physics-based, data-driven, and hybrid modeling paradigms in condition monitoring and predictive maintenance.

Table 2: Taxonomy of modeling approaches commonly adopted in condition monitoring and predictive maintenance, synthesized from recent surveys on physics-based, data-driven, and physics-informed machine learning models [23, 24].

Category	Typical formulations / examples
Physics-Based Models	First-principles models; analytical / white-box models; simulation-based models (Finite Element Method (FEM) / Computational Fluid Dynamics (CFD)).
Data-Driven Models	Machine learning: classical machine learning (ML); deep learning including Multilayer Perceptrons (MLP), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Transformers, and Graph Neural Networks (GNN).
Hybrid / Physics-Informed Models	Grey-box models; Physics-Informed Neural Networks (PINNs); physics-guided deep learning; causality- or physics-guided Graph Neural Networks (GNNs).

Table 3: Machine learning paradigms and representative model families, summarizing commonly used architectures, inductive biases, and data modalities as reported in recent surveys on deep learning, predictive maintenance, multi-objective learning, and graph-based spatio-temporal modeling [24–27].

Learning paradigm	Model family / representative examples	Inductive bias (summary)	Typical data
Supervised	Classical ML: linear and logistic regression; Support Vector Machines (SVM); k-Nearest Neighbors (kNN); decision trees; random forests; gradient boosting	Linearity; margin maximization; locality in feature space; rule-based partitioning; ensemble averaging	Labeled tabular features; engineered signals
	Deep Learning: Multilayer Perceptrons (MLP); Convolutional Neural Networks (CNN); Recurrent Neural Networks (RNN), including Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU); Transformers (including encoder–decoder architectures)	Representation learning; spatial locality (CNN); temporal order (RNN); global attention mechanisms (Transformers)	Images; time series; sequences
	Graph learning: Graph Neural Networks (GNNs) and graph transformers	Relational and topological inductive bias (message passing or attention over graphs)	Graph-structured systems; networks
Unsupervised	Classical ML: k-means clustering; hierarchical clustering; Principal Component Analysis (PCA)	Cluster geometry assumptions; nested partitions; linear subspace representations	Unlabeled features; embeddings
	Deep representation / generative learning: autoencoders (sparse and denoising); Variational Autoencoders (VAE); diffusion models	Compressible latent representations; reconstruction-based learning; probabilistic latent spaces; iterative denoising generative priors	High-dimensional signals; anomaly detection; generative modeling
Self-supervised	Graph learning: unsupervised or generative graph models (via autoencoders, contrastive learning, or diffusion)	Relational representations without labels; structural priors derived from graph topology	Unlabeled graphs; topology with measurements
	Representation learning: contrastive learning (e.g., Simple Framework for Contrastive Learning of Visual Representations (SimCLR), Momentum Contrast (MoCo)); masked prediction (e.g., Masked Autoencoders (MAE), Masked Language Modeling (MLM)); self-distillation	Learning invariances and context completion via pretext tasks; scalable representation learning	Large unlabeled datasets (images, sensor signals, text)
	Foundation models: Transformers and Large Language Models (LLMs) (pretraining followed by task-specific adaptation)	Attention-based sequence modeling; broad statistical priors acquired through large-scale pretraining	Text and code (LLMs); multimodal corpora; tokenized signals
Reinforcement learning	Graph self-supervised learning: contrastive or masked objectives on graphs; graph transformers	Topology-aware relational representation learning without explicit labels	Large graph datasets; relational sensor graphs
	Classical RL: Q-learning; State–Action–Reward–State–Action (SARSA)	Markov Decision Process (MDP) assumption; value-based learning in discrete state–action spaces	Interactive environments; low-dimensional control
	Deep RL: Deep Q-Networks (DQN); policy gradient methods; actor–critic methods	Neural function approximation; exploration versus exploitation; learned state representations	High-dimensional observations; robotics and control
	RL for foundation models: preference optimization and Reinforcement Learning from Human Feedback (RLHF) pipelines	Reward modeling and alignment via preferences; policy optimization in output space	Human feedback; instruction data

2.4 Hydraulic fault diagnosis under multirate sampling

Hydraulic systems typically involve heterogeneous sensors operating at different sampling rates, complicating temporal alignment and end-to-end learning. Huang *et al.* propose a deep learning framework explicitly designed for multirate hydraulic data, achieving robust diagnostic accuracy even under severe sampling imbalance and class skew [28]. Complementary multirate diagnostic architectures integrate efficient convolution variants and feature fusion mechanisms: Qu *et al.* combine residual shrinkage separable CNNs with statistical features and gated attention fusion to aggregate high- and low-rate information [29], while Zhang *et al.* employ fully convolutional networks for multi-fault diagnosis from multirate time series [30]. Beyond deep learning, Zheng *et al.* introduce a total multirate linear Gaussian state-space model that separates global dynamic latent variables from rate-specific static components, enabling statistically grounded fault detection across sampling regimes [31]. These contributions motivate the multi-resolution data backbone adopted in this paper, combining high-frequency edge acquisition with aggregated analytical views suitable for scalable learning.

From an industrial perspective, recent experimental evidence highlights that hydraulic degradation is tightly linked to specific operating regimes rather than isolated fault events. In particular, load-holding operation has been shown to accelerate wear progression in axial piston pumps, resulting in increased internal leakage and altered pressure dynamics, which can serve as reliable health indicators for data-driven prognostics [32].

2.5 RUL estimation under limited and heterogeneous degradation data

RUL prediction remains challenging due to scarce run-to-failure datasets and heterogeneous operating conditions. Health-index-based approaches provide a practical abstraction: Liu *et al.* propose RUL estimation via health index similarity using distance and directional metrics [33], while Yang *et al.* introduce dynamic smoothing and ensemble realization to capture nonlinear degradation trends [34]. To address domain shift and incomplete degradation trajectories, Li *et al.* propose partial domain adaptation using adversarial learning when only early target-domain degradation data are available [35], and Cheng *et al.* combine dynamic modeling with transfer learning and generative data augmentation to mitigate the lack of full-cycle degradation data [36]. More recent work reframes prognostics by modeling degradation trajectories directly: Fan *et al.* propose Degradation Path Approximation (DPA), deriving RUL from aligned health indicator trajectories under limited historical data [37]. However, such formulations typically assume continuous and monotonic degradation, whereas industrial electro-hydraulic systems often exhibit regime changes and economically non-viable operating states rather than smooth failure trajectories. This motivates alternative RUL formulations grounded in state transitions and operational thresholds, as adopted in the present work.

2.6 Architectural perspectives: streaming, data management, and scalability

Despite algorithmic maturity, scalable PHM deployment critically depends on data ingestion, streaming, and storage architectures. Kappa-style streaming architectures have been shown to reduce latency and complexity compared to hybrid Lambda designs in continuous real-time analytics [38, 39]. Scalable stream aggregation and fault-tolerant sensor analytics are further addressed by Henning and Hasselbring, who demonstrate reliable multidimensional aggregation in Industry 4.0 data streams [40]. At the persistence layer, comparative studies show that NoSQL and time-series databases are well suited for high-rate industrial data ingestion [41, 42], while relational and warehouse-oriented systems remain advantageous for structured metadata, cross-site reporting, and analytical traceability [43, 44]. Recent surveys on data ingestion pipelines and IIoT architectures further highlight the importance of edge-cloud partitioning, protocol interoperability, and scalable ETL design for industrial AI systems [45–47].

2.7 Positioning of this work

Much of the existing literature on fault diagnosis and RUL estimation emphasizes increasingly sophisticated machine learning and deep learning models, often assuming the availability of degradation labels or health indicators that are consistent and physically meaningful. In practical hydraulic systems, this assumption rarely holds.

Hydraulic degradation typically manifests through discrete operational events, such as abnormal pressure regimes, temperature-induced slowdowns, or system downtime driven by economic or maintenance-related constraints, rather than as smooth numerical trajectories. Without a structured record of faults, stoppages, and maintenance activities, RUL predictions, regardless of model complexity, remain largely referential and fail to capture their full industrial applicability.

While several frameworks propose end-to-end predictive maintenance pipelines, data acquisition and event logging are frequently treated as secondary implementation aspects. As a result, the connection between model outputs and the underlying physical and operational causes of degradation remains weak, limiting both interpretability and economic value.

The present work is positioned on the premise that meaningful and actionable RUL estimation becomes possible when data-driven models are supported by sustainable data acquisition and structured records of operational events. Using a real-world electro-hydraulic case study controlled by an industrial PLC, this work demonstrates how fault and maintenance records provide the necessary context to interpret model outputs and extract practical value in hydraulic systems.

3 Hydraulic System Description

For this application case, we analyze a hydraulic system whose core components are listed in Table 4 and its hydraulic diagram is depicted in Figure 1. This system is designed to simulate a hydraulic press.

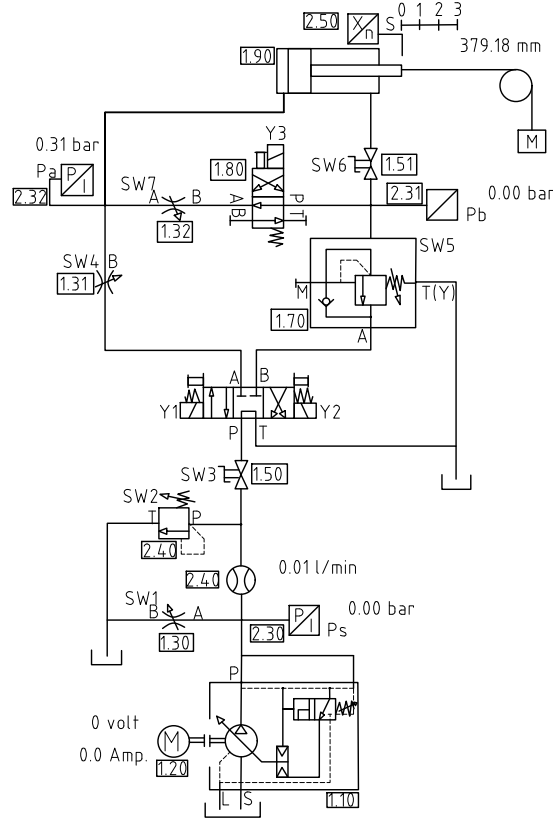


Fig. 1: Hydraulic drawing of the system.

Table 4: Rexroth components list.

Item	Quantity	Designation	Type	Rexroth material num.
1.10	1	Power Unit	XITE Hydraulix 300 workstation	R961009456
2.40	1	Gear wheel flow rate sensor	TS-HC-GFM 3843-03-S-3	R961002508
1.30-31-32	3	Throttle valve	TS-HC-DV 06-1-1X/V	R961002539
1.40	1	Pressure relief valve	TS-HC-DBDH 6 G1X/100&	R961002520
1.60	1	4/3 directional control valve	TS-HC-4WE 6 G6X/EG24&	R961002548
1.70	1	Pressure sequence valve	TS-HC-DZ 6 DP1-5X/75Y	R961002558
1.80	1	4/2 directional control valve	TS-HC-4WE 6 C6X/EG24&	R961002547
1.90	1	Load unit, 80 kg (stationary)	TS-HC-CD 70H 25/16-&	R961004005
2.30-31-32	3	Pressure sensor	TS-HC-3403-17-S-E5.37	R901465414
2.50	1	Position sensor	-	-
-	3	Hydraulic distributor (4 ports)	TS-HC-4F-3M1W	R961002485

Note: Component data are taken from Bosch Rexroth specification sheets [48].

The system is supervised by a PLC that executes standard threshold-based logic for operational safety. For this study, the PLC is interfaced with a ctrlX CORE platform. The integration of an OPC UA server on this controller is pivotal to the proposed

dataflow framework, enabling the high-frequency acquisition of time-series sensor data. This data is subsequently stored in a structured database and serves as the input for the prediction pipelines developed for anomaly detection and prognostics.

For our analysis, we assume a continuous extension and retraction cycle of the cylinder. Within this operational context, we specifically investigate two primary failure modes:

- **Excessive Internal Leakage:** Leading to cylinder stoppage due to significant internal fluid loss.
- **Anomalous Operation (Slowdown):** Characterized by a sudden increase in oil temperature, causing a deceleration in cylinder extension and retraction.

The displayed hydraulic distribution allows for the simulation of internal cylinder leakage, specifically between the high and low-pressure zones, by activating a valve controlled by relay Y3 as shown in Figure 1. Critically, this system intentionally does not include a cooling mechanism. Consequently, over time, the oil temperature gradually increases, leading to anomalous operating conditions. This situation is further exacerbated when internal cylinder leakage is simulated, often culminating in the complete cessation of cylinder movement under critical conditions.

This foundational setup, established through the construction of fault detection tables (Section 4.1), enabled the identification of potential system failures. Moreover, it defined the forward path for implementing predictive models specifically aimed at mitigating two critical failure modes: excessive internal leakage and anomalous operation (slowdown). These models integrate the corresponding physical variables and incorporate insights from hydraulic experts.

For safety reasons in an controlled commissioning and training environment, and despite the use of real hydraulic system components and configuration, a cooling system was intentionally not installed. The primary objective of this setup is didactic and educational.

3.1 Mathematical Model and PLC-Oriented Control Abstraction

Table 5 summarizes the symbols, units, and signal representations used in the hydraulic-cylinder model and subsequent equations

The hydraulic actuator is controlled by a PLC through discrete valve commands, resulting in a hybrid continuous–discrete control structure. The PLC does not directly implement a continuous controller, but instead selects discrete valve states that excite the underlying continuous hydraulic dynamics. For this reason, the physical system is modeled as a continuous-time plant driven by a discrete-valued control input.

The plant dynamics follow well-established electro-hydraulic actuator formulations, including force balance, chamber continuity, and orifice-based valve flow relations [8, 9, 11].

The cylinder kinematics are given by

$$\dot{X}_{cyl} = V_{cyl}. \quad (1)$$

Table 5: Nomenclature aligned with measured signals and the hydraulic-cylinder model

Symbol	Unit	Meaning and relation to measured signals
X_{cyl}	mm	Cylinder position (measured position sensor); model state in meters.
$V_{\text{cyl}} = \dot{X}_{\text{cyl}}$	mm/s	Cylinder velocity (measured); model state in meters per second.
V_{ret}	mm/s	Retraction velocity feature, e.g. $V_{\text{ret}} = \min(V_{\text{cyl}}, 0)$.
m	kg	Equivalent moving mass (piston, rod, and load).
P_A	bar	Pressure at port/chamber A (extension side).
P_B	bar	Pressure at port/chamber B (retraction side).
P_S	bar	Supply pressure (HPU / supply line).
P_T	bar	Tank/return pressure (often near constant).
A_A	m ²	Effective piston area, chamber A.
A_B	m ²	Effective piston area, chamber B.
$V_A(X_{\text{cyl}})$	m ³	Fluid volume in chamber A as a function of piston position.
$V_B(X_{\text{cyl}})$	m ³	Fluid volume in chamber B as a function of piston position.
Q_A	m ³ /s	Flow into chamber A.
Q_B	m ³ /s	Flow into chamber B.
Q_{oil}	l/min	Oil flow-rate signal; optional mapped feature from model flows.
$\beta_e(T)$	Pa	Effective bulk modulus (temperature dependent).
ρ	kg/m ³	Hydraulic oil density.
$F_{\text{fric}}(V_{\text{cyl}}, T)$	N	Friction force (velocity and temperature dependent).
$F_{\text{load}}(X_{\text{cyl}}, t)$	N	External load force acting on the actuator.
$C_d(T)$	–	Valve discharge coefficient (temperature dependent).
$A_v(u)$	m ²	Effective valve orifice area as a function of PLC command u .
$C_\ell(T)$	m ³ /s/Pa	Internal leakage coefficient (temperature dependent).
u	–	Discrete PLC valve command: $u \in \{-1, 0, +1\}$ (retract / stop / extend).
T	°C	Hydraulic oil temperature.
$P_{A/S}$	–	Pressure ratio feature $P_{A/S} = P_A/P_S$.
$P_{B/A}$	–	Pressure ratio feature $P_{B/A} = P_B/P_A$.

Note: The physical model is typically implemented in SI units (m, s, Pa). Measured signals may be stored in engineering units (mm, mm/s, bar, l/min); consistent unit conversion is applied during identification and machine-learning preprocessing.

The translational dynamics follow from Newton’s second law:

$$m \dot{V}_{\text{cyl}} = A_A P_A - A_B P_B - F_{\text{fric}}(V_{\text{cyl}}, T) - F_{\text{load}}(X_{\text{cyl}}, t). \quad (2)$$

A compact friction representation is adopted:

$$F_{\text{fric}}(V_{\text{cyl}}, T) = F_c(T) \text{sgn}(V_{\text{cyl}}) + b(T) V_{\text{cyl}}. \quad (3)$$

For consistency with the feature set used in data-driven analysis, the retraction velocity is defined as

$$V_{\text{ret}} = \min(V_{\text{cyl}}, 0). \quad (4)$$

3.1.1 Hydraulic Pressure Dynamics

The chamber pressure dynamics follow continuity:

$$\dot{P}_A = \frac{\beta_e(T)}{V_A(X_{\text{cyl}})} \left(Q_A - A_A V_{\text{cyl}} - C_\ell(T) (P_A - P_B) \right), \quad (5)$$

$$\dot{P}_B = \frac{\beta_e(T)}{V_B(X_{\text{cyl}})} \left(Q_B + A_B V_{\text{cyl}} - C_\ell(T) (P_B - P_A) \right). \quad (6)$$

The chamber volumes depend on piston position:

$$\begin{aligned} V_A(X_{\text{cyl}}) &= V_{A0} + A_A X_{\text{cyl}}, \\ V_B(X_{\text{cyl}}) &= V_{B0} + A_B (L - X_{\text{cyl}}). \end{aligned} \quad (7)$$

3.1.2 Valve Flow Model and PLC Interface

The directional valve is actuated by a PLC with discrete command

$$u \in \{-1, 0, +1\}, \quad (8)$$

where $u = +1$ denotes extension, $u = -1$ retraction, and $u = 0$ a blocked/hold state.

A minimal orifice-based mapping from discrete logic to continuous flows is

$$\begin{aligned} Q_A &= C_d(T) A_v(u) \sqrt{\frac{2}{\rho}} \\ &\quad \times \sqrt{\max(P_S - P_A, 0)}, \end{aligned} \quad (9)$$

$$\begin{aligned} Q_B &= - C_d(T) A_v(u) \sqrt{\frac{2}{\rho}} \\ &\quad \times \sqrt{\max(P_B - P_T, 0)}. \end{aligned} \quad (10)$$

To match the plotted features used for condition monitoring, we additionally define

$$P_{A/S} = \frac{P_A}{P_S}, \quad P_{B/A} = \frac{P_B}{P_A}. \quad (11)$$

The flow-rate signal is reported as a derived channel:

$$Q_{\text{oil}} \triangleq \alpha |Q_A| \quad (\text{or } \alpha |Q_B|), \quad (12)$$

This notation ensures that (i) the physics-based states map directly to measured channels $(X_{\text{cyl}}, V_{\text{cyl}}, P_A, P_B, P_S)$ and (ii) the PLC-level discrete decision u is explicitly

connected to the continuous hydraulic dynamics used for subsequent identification and ML.

Remark on PLC Signal Conditioning

Measured signals are subject to standard PLC scaling and low-pass filtering, which can be approximated by first-order (PT1) dynamics and are therefore not modeled explicitly.

4 Methods

This section presents the main architectural components required to support a predictive maintenance solution aligned with Industry 4.0 principles:

- A database architecture for structured recording of machine faults and downtime events.
- A data flow architecture (ETL pipeline) to collect, process, and store real-time sensor data from PLCs into time-series tables.

Together, these architectures provide a unified data backbone for implementing anomaly detection and RUL prediction models.

In industrial environments, anticipating equipment degradation provides critical lead time for proactive maintenance planning, enabling a transition from reactive interventions to scheduled and economically optimized actions. However, the practical deployment of prognostic models is often hindered by the scarcity of complete run-to-failure datasets, as catastrophic breakdowns are rare, safety-critical, and actively avoided in real systems.

To address this limitation, RUL is redefined in this work not as the time to physical component failure, but as the expected time until the system transitions into an operational state that is no longer economically viable. These operational states are identified through a data-driven state model based on K-Means clustering and HMMs, and are validated using expert knowledge and structured downtime records.

Accordingly, RUL prediction is formulated as a state-transition forecasting problem, where the target variable represents the remaining time until a predefined degradation or critical operating state is reached, rather than the extrapolation of a continuous degradation trajectory.

4.1 Database Architecture for Fault Collection

A minimal yet standardized relational schema is proposed to support the automatic recording of machine faults and downtimes in a structured and query efficient manner. The design enables both consistent taxonomy for supervised learning and traceable performance monitoring aligned with Industry 4.0 principles. When a non-operational state is detected—via the `run_status` signal derived from sensor thresholds or PLC logic—an event is automatically logged into the database. A graphical interface allows operators to validate and enrich these records with contextual details (e.g., fault codes, failure causes, or maintenance notes), addressing limitations of purely sensor-driven diagnosis.

Importantly, the proposed database is not designed as a repository of catastrophic failure events, which are typically scarce in industrial practice, but as a structured entry point for exploratory analysis. It captures equipment stoppages and non-operational states together with their operationally defined reasons, even when detailed fault mechanisms are not yet fully identified. This event-centric representation establishes the initial operational context required to relate time-series sensor data to system behavior, and provides a foundation for subsequent physical interpretation and deeper diagnostic analysis when finer-grained failure information becomes available.

The event logging process, illustrated in Figure 2, includes the following components:

- **Location_unit**: hierarchical mapping of plant areas.
- **Asset**: inventory of hydraulic equipment linked to locations.
- **Calendar_dim**: temporal dimension for aligning events with production schedules.
- **Event_log**: central table recording downtime events with asset IDs, timestamps, and related categories, problems, and causes.
- **Event_category**, **Problem_type**, **Root_cause**: classification tables supporting consistent fault taxonomy.

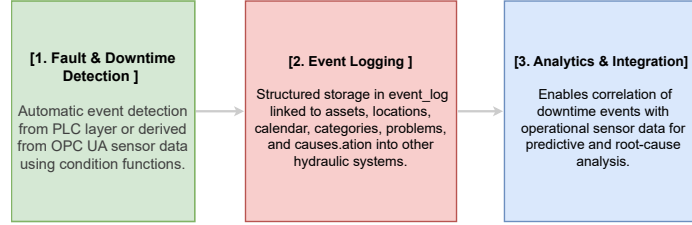


Fig. 2: Standard data collection behavior within the proposed framework. This diagram illustrates the normal acquisition and logging process for operational and fault events. For the detailed database organization and schema design of the logging system, refer to Appendix B.

Table 6: Downtime Category Definitions

Category	Definition
Operational Standby	Equipment is available but idle due to external factors such as lack of upstream input or scheduling constraints.
Planned Maintenance	Scheduled inspections, calibration procedures, or preventive part replacement activities.
Unplanned Failure (Emergency)	Sudden production stop caused by unexpected system failures; in hydraulic systems this may include pressure loss, valve malfunction, or actuator blockage.
Resource Unavailability	Downtime caused by missing or insufficient resources, such as low oil levels, blocked supply lines, or unavailable utilities in hydraulic systems.

Note: The categorization follows common industrial downtime classification schemes and is adapted to hydraulic and fluid power systems where applicable.

Downtime events are categorized according to predefined operational taxonomies and associated with contextual sensor data (e.g., pressure, stroke length, temperature). Rather than constituting a physical failure ground truth, these records provide an objective and automatically generated operational reference layer that captures why and under which conditions a machine transitioned into a non-operational state.

This first-layer event classification enables a clear separation between operational standby, planned maintenance, unplanned equipment-related stoppages, and external resource or upstream process unavailability. By explicitly distinguishing machine-internal issues from line-level or organizational constraints, the database establishes a reliable baseline for subsequent exploratory analysis and data-driven modeling.

The resulting structured dataset (Table 7) supports systematic operational root-cause *investigation* and the training of predictive models, without requiring explicit identification of detailed physical failure mechanisms. Post-deployment, it further enables objective model evaluation and continuous monitoring of maintenance and availability KPIs, serving as a foundational enabler for scalable and industrially applicable prognostic solutions.

Table 7: Hydraulic System Event Log

ID	Asset	Start	Stop	Dur.	Shift	Date	Comment	Cat.	Prob.	Cause	Sev.	RA
1	AST001	10:10	10:40	30	Day	25-07-30	Cylinder not extending/retracting; inter-nal pressure drop.	3	101	501	7.5	H
2	AST001	13:20	13:50	30	Day	25-07-31	Extension speed unusually slow; likely pump inefficiency.	3	102	502	6.8	M
3	AST003	15:00	15:10	10	Night	25-07-29	Routine inspection of oil cooler; no anomalies found.	2	0	0	2.0	O
4	AST002	08:00	09:15	75	Day	25-08-01	Pressure accumulator failed to hold charge.	3	103	503	8.2	H
5	AST004	22:00	22:45	45	Night	25-08-02	Leak detected at main hose connection.	3	104	504	7.0	M
6	AST001	11:30	12:00	30	Day	25-08-03	Filter blockage indicated by high differential pressure.	3	105	505	6.5	O
7	AST005	06:00	06:15	15	Night	25-08-04	Sensor calibration required.	2	0	0	3.0	I

Note: Duration is given in minutes. Severity score ranges from 1 (minor) to 10 (critical). RA denotes the responsible area: H—Hydraulics, M—Maintenance, O—Operations, I—Instrumentation.

4.2 ETL Architecture for Sensor Data Acquisition

The proposed ETL (Extract, Transform, Load) framework, whose architecture is depicted in Figure 3, enables continuous acquisition of time-series sensor data from OPC UA-enabled devices. Raw measurements are ingested in real time, transformed into a standardized format, and stored in normalized relational tables suitable for querying, visualization, and machine learning.

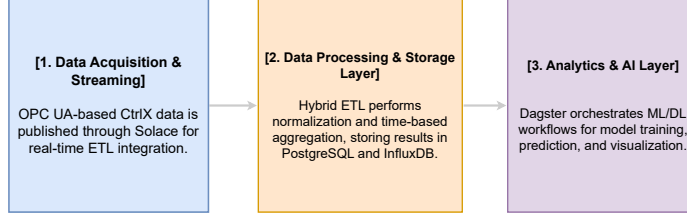


Fig. 3: Conceptual overview of the data streaming and ETL architecture underlying the proposed research framework. The diagram highlights the main stages from acquisition to storage and analytics. A detailed orchestration scheme is provided in Appendix B.

As described in Algorithms 1, 2, and 3, all components rely on open-source technologies, ensuring adaptability and eliminating licensing constraints. The messaging backbone is built upon Solace PubSub+, available in its Standard Edition at no cost [49].

4.2.1 Rationale for Hybrid Time-Series and Relational Storage

While time-series databases such as InfluxDB are well suited for high-frequency sensor ingestion and exploratory visualization, their variable-centric dashboards (e.g., Grafana panels per signal) provide limited support for holistic asset-level reasoning when the number of signals and monitored assets scales. In large-scale industrial deployments, this often results in fragmented insights that hinder rapid assessment and decision-making by end users.

To address this limitation, the proposed architecture deliberately combines a non-relational time-series store with a relational data model derived through scheduled pivoting operations, as formalized in Algorithms 1, 2 and 3. Raw measurements are preserved in InfluxDB to retain full temporal fidelity, while frequency-aligned relational tables are generated to enable efficient aggregation, reporting, and machine-learning workflows.

It is emphasized that the selected technologies represent one possible realization rather than the only viable implementation. Messaging topics, scheduling mechanisms, and orchestration layers may be instantiated through alternative solutions depending on deployment constraints. The architectural principle—decoupling raw data fidelity from analysis-ready representations—remains invariant.

This work emphasizes a transparent and interpretable data engineering methodology that balances computational efficiency with analytical value. By explicitly structuring

data transformations and aggregation layers, the proposed approach avoids treating industrial analytics as a black box and enables domain experts to relate aggregated indicators to underlying physical phenomena.

Beyond its infrastructural role, the proposed ETL framework is designed to explicitly support degradation-aware learning. The multi-resolution aggregation strategy preserves fast transient phenomena at high sampling rates while providing statistically stable representations at coarser scales, which are required for reliable RUL estimation. This separation enables prognostic models to disentangle short-term dynamics from long-term wear evolution without loss of physical interpretability.

Algorithm 1 Edge-Oriented ETL Framework for PLC-Integrated Hydraulic Systems
(Part 1: Initialization & Extraction)

Require: Real-time sensor data from `ctrlX CORE` (via OPC UA)

Ensure: Pivoted production tables in PostgreSQL and a cloud-ready data lake

Phase 1: Service Initialization

- 1: Start Docker containers:
 - PostgreSQL with `pg_cron` extension
 - Solace Broker (MQTT)
 - InfluxDB time-series database
 - Telegraf metrics collector
 - Dagster Orchestrator
 - ETL Microservices (Extract, Transform, Load)
 - Grafana visualization platform

Phase 2: Data Extraction

- 2: Establish OPC UA connection
 - 3: **if** simulation mode is active **then**
 - 4: Connect ETL-Extract to the simulated CtrlX server
 - 5: **else**
 - 6: Connect to the physical CtrlX OPC UA server
 - 7: **end if**
 - 8: Publish raw sensor readings:
 - Topic: `opcua/data`
 - Format: `{sensor_id, value, timestamp}`
-

Algorithm 2 Edge-Oriented ETL Framework (Part 2: Transformation & Loading)

Phase 3: Data Transformation

- 1: Subscribe to the `opcua/data` topic
 - Apply value scaling (e.g., 0-100% to engineering units)
- 2: Publish processed data
 - Topic: `ctrlx/transformed`
 - Add quality flags to the payload

Phase 4: Data Loading

- 3: Perform batch processing
 - Buffer incoming records in 5-minute windows
 - Bulk insert into the PostgreSQL long table
 - Schema: `(sensor_id, value, timestamp, quality)`
-

Algorithm 3 Edge-Oriented ETL Framework (Part 3: Pivoting & Monitoring)

Phase 5: Data Pivoting

- 1: Schedule pivoting jobs
 - `pg_cron` triggers at 100ms, 1s, and 1min intervals
 - Generate frequency-synchronized tables

Phase 6: Orchestration & Monitoring

- 2: Provide system oversight
 - Dagster monitors service health
 - Automatic restart on failure detection
 - 3: Collect metrics
 - Use Telegraf → InfluxDB → Grafana pipeline
 - Track latency, throughput, and error rates
 - 4: Perform optional cloud integration
 - Parallel write to a data lake
 - Enable scalable analytics
-

Table 8 presents a representative data output with a 1-minute temporal resolution. In practice, the pipeline maintains three parallel output tables corresponding to aggregation frequencies of 100 ms, 1 s, and 1 min. These are refreshed every 5 minutes to consolidate recent activity into structures optimized for downstream processing.

To preserve diagnostic fidelity, raw acquisition is conducted at 100 ms intervals—essential for capturing fast transients such as internal leakage, stick-slip events, or abrupt pressure changes. These phenomena are often imperceptible at coarser resolutions but are critical indicators of early-stage degradation relevant for prognostic models such as RUL estimators.

Importantly, the architecture is vendor-agnostic and fully generalizable to any industrial machine equipped with an OPC UA server. Its configurable signal mapping and multi-resolution design ensure scalability across a wide range of applications, from high-speed hydraulic systems to slower electromechanical assets.

In summary, the ETL framework acts as a foundational enabler for scalable, high-fidelity prognostics by bridging low-level sensor acquisition with structured, analysis-ready data representations.

Table 8: Example output table with 1-minute frequency.

Timestamp	ENER	ACT_VE	EFF_SY	p_A
2025-05-27 11:37:00	225.88	0.60	21.99	1.87
2025-05-27 11:35:00	225.38	-2.45	27.04	1.98
2025-05-27 11:34:00	225.23	1.20	35.30	0.07
2025-05-27 11:33:00	225.43	-4.65	27.99	2.05
2025-05-27 11:32:00	225.28	-4.75	31.84	2.04
2025-05-27 11:31:00	225.38	-5.38	30.77	2.12

Note: Similar structures apply for 100 ms and 1-second intervals.

5 Machine and Deep Learning

ML and DL have emerged as transformative tools for supporting operational decision-making in industrial environments. By integrating real-time and historical sensor data,

these approaches enable rapid visualization of system states and facilitate data-driven intervention strategies. However, their value must be rigorously measured—not only in terms of technical performance (e.g., accuracy, robustness) but also through broader economic impact indicators such as cost savings, operational efficiency gains, and enhanced system reliability.

Successful deployment of ML/DL solutions requires meeting three critical conditions:

1. **Model Performance:** Achieve statistically robust accuracy and stability under varying operational conditions.
2. **Domain Alignment:** Ensure model outputs are consistent with expert knowledge and physically meaningful system behavior.
3. **Stakeholder Adoption:** Gain full acceptance from domain experts and decision-makers by providing interpretable and operationally relevant results.

5.1 Methodological Contribution and Impact

This work proposes a hybrid anomaly-to-failure progression modeling framework that leverages unsupervised state discovery, where system states are inferred directly from measurement data without requiring labeled examples, combined with domain-informed interpretation.

Due to the absence of explicit state or degradation labels during normal operation, an unsupervised learning approach is employed to cluster operational data into latent system states based solely on observed measurements. These data-driven states are subsequently interpreted and labeled a posteriori by correlating their temporal evolution with logged failure events and known hydraulic system behavior.

In the first stage, unsupervised algorithms such as K-Means clustering and HMMs identify latent operational states from hydraulic system data. A key innovation is the *expert annotation* of cluster centroids and state signatures, linking abstract data patterns to physical phenomena such as hydraulic pressure thresholds, leakage indicators, or temperature deviations.

Before RUL estimation, the framework performs a *temporal correlation analysis* between operational time and state-transition sequences (e.g., normal $\rightarrow$ critical $\rightarrow$ highly critical).

After this correlation stage is completed, the RUL modeling is conducted:

- **RUL:** Estimated time until the system transitions from normal to critical operation.
- **Critical RUL:** Remaining time before the onset of high-risk operational conditions.

Unlike conventional RUL estimation, which relies exclusively on rare failure-event datasets, this framework focuses on precursor states validated by experts, thereby eliminating the dependency on explicit fault data. Nevertheless, if actual failure events are available, they are naturally incorporated into the training process to refine the estimation of RUL boundaries. This method enables intervention as soon as early deviations from nominal operation are detected, while still benefiting from the additional accuracy provided by real-world fault cases.

5.2 Data Description

Figure 4 illustrates the empirical distributions of the raw variables used in this study. The preprocessing steps applied to each signal are detailed in Sections 5.4 and 5.3.

To prevent bias in the statistical characterization caused by sensor dropouts, interpolation artifacts, or physically implausible actuator behavior, the raw measurements are processed through a structured data cleaning pipeline, summarized in Algorithm 4. The objective of this procedure is not aggressive denoising, but the preservation of physically meaningful dynamics while excluding invalid samples, thereby reducing the risk of artificial degradation trends caused by interpolation artifacts or physically implausible actuator behavior.

Algorithm 4 Data Cleaning Process (Class `DataCleaner`)

- 1: Load configuration parameters (NaN thresholds, filter windows)
 - 2: Interpolate missing values for signals below NaN ratio threshold
 - 3: Clean velocity signal (`CtrlX_2_ACT_VE`):
 interpolation, smoothing, and detection of abnormal peaks
 - 4: Clean position signal (`CtrlX_2_ACT_P0`):
 outlier removal and smoothing
 - 5: Identify and remove invalid samples based on pressure constraints
-

For both classification and RUL regression tasks, all subsequent analyses operate on the cleaned data. In particular, the cylinder retraction velocity is emphasized, as this operating phase reveals the effects of excessive internal leakage and anomalous slowdowns, as discussed in Section 3. Likewise, the pressure ratios provide direct insight into leakage-related system behavior, as recommended by hydraulic experts.

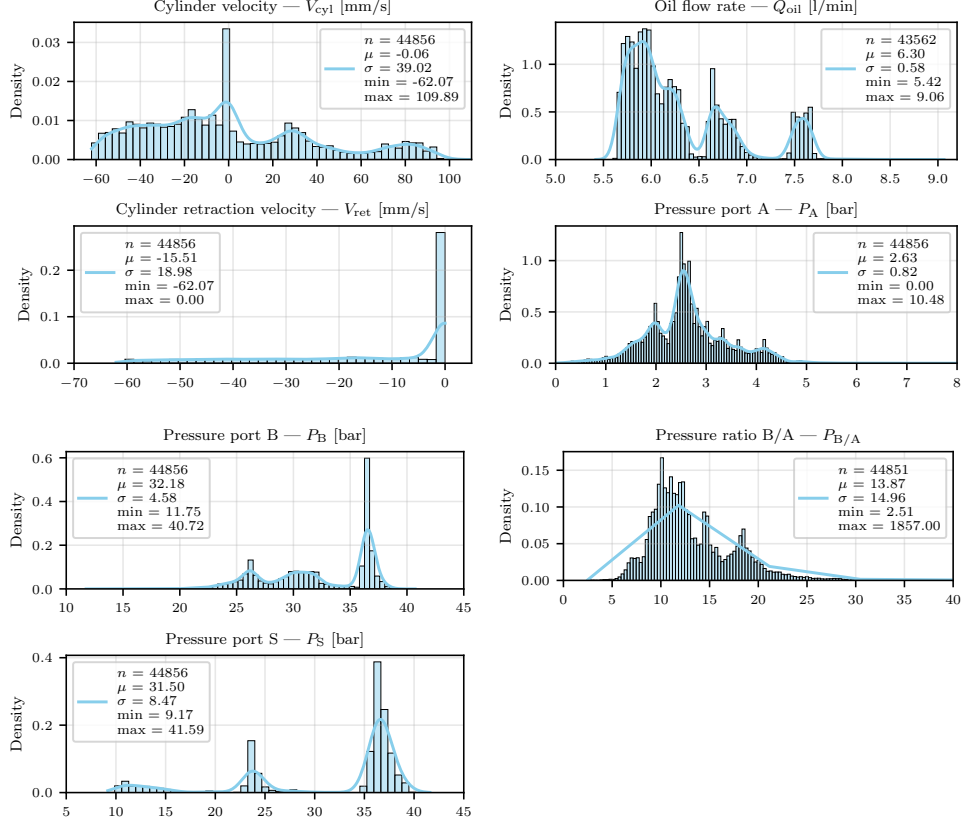


Fig. 4: Empirical distributions of raw variables, including histograms, KDE, and statistical summaries across four variable sets.

5.3 Machine State Classification

For condition monitoring and anomaly detection, the unsupervised clustering algorithm K-Means was employed to identify abnormal patterns. According to Chandola et al. [50], clustering-based methods are widely recognized as effective techniques for anomaly detection. Building on this foundation, Hidden Markov Models (HMM) were additionally applied to detect operational states. These models were supported by preprocessing techniques that calculated signal variation, oscillation, and envelope characteristics over time to distinguish critical operating states.

5.3.1 Preprocessing

Data preprocessing followed the methodology outlined in Algorithm 5. One of the key challenges, as observed in the ETL output table (Section 4.2), is the high-frequency

variability inherent to hydraulic systems. Due to their fast dynamic response, variables such as pressure, position, and flow rate can exhibit significant changes within milliseconds — for instance, cylinder velocity, expressed in mm/s, fluctuates rapidly.

From a physical perspective, hydraulic degradation manifests primarily through changes in the consistency and repeatability of motion rather than absolute signal levels. In short-term observation windows, these effects are more accurately described as performance degradation, reflecting a loss of dynamic efficiency rather than irreversible mechanical wear.

Internal leakage and viscosity reduction alter the force balance within the cylinder, leading to increased cycle-to-cycle variability in both position and velocity signals. As degradation progresses, these effects appear as (i) gradual widening of the signal envelopes, (ii) increased oscillation variance within each stroke, and (iii) a progressive increase in the cycle period due to reduced effective actuation efficiency.

Envelope-based representations therefore provide a compact means of capturing both transient performance degradation and longer-term trends, while oscillation and cycle-based features quantify short-term dynamic irregularities that may precede or accompany hydraulic wear.

Algorithm 5 Feature Engineering Process (Class `DataProcessor`)

- 1: Compute oscillation metrics from cleaned position signal
 - 2: Detect position peaks and extract cycle-based amplitude and period
 - 3: Compute rolling statistics (mean, std) of amplitude and period features
 - 4: Estimate velocity and position envelopes and apply smoothing
 - 5: Suppress amplitude and period features outside inverse velocity peak regions
-

To capture these dynamics adequately, a sampling frequency of 100 ms was selected. This temporal resolution proved critical for detecting early signs of system degradation, such as internal leakage. Notably, the velocity signal displays localized anomalies — small amplitude increases — which become distinguishable only at higher sampling rates (Figure 6). These patterns provide valuable information for predictive models, including those used for (RUL estimation via XGBoost and LSTM with attention mechanisms).

Figures 5 present derived features analyzed within a moving window framework. Each 'cycle' corresponds to a full stroke of the hydraulic cylinder. Over time, both the position signal and its derivative (velocity) exhibit varying amplitude and period, which are indicative of evolving mechanical degradation. These variations stem primarily from internal leakage and dynamic limitations that affect the system's ability to maintain consistent motion profiles.

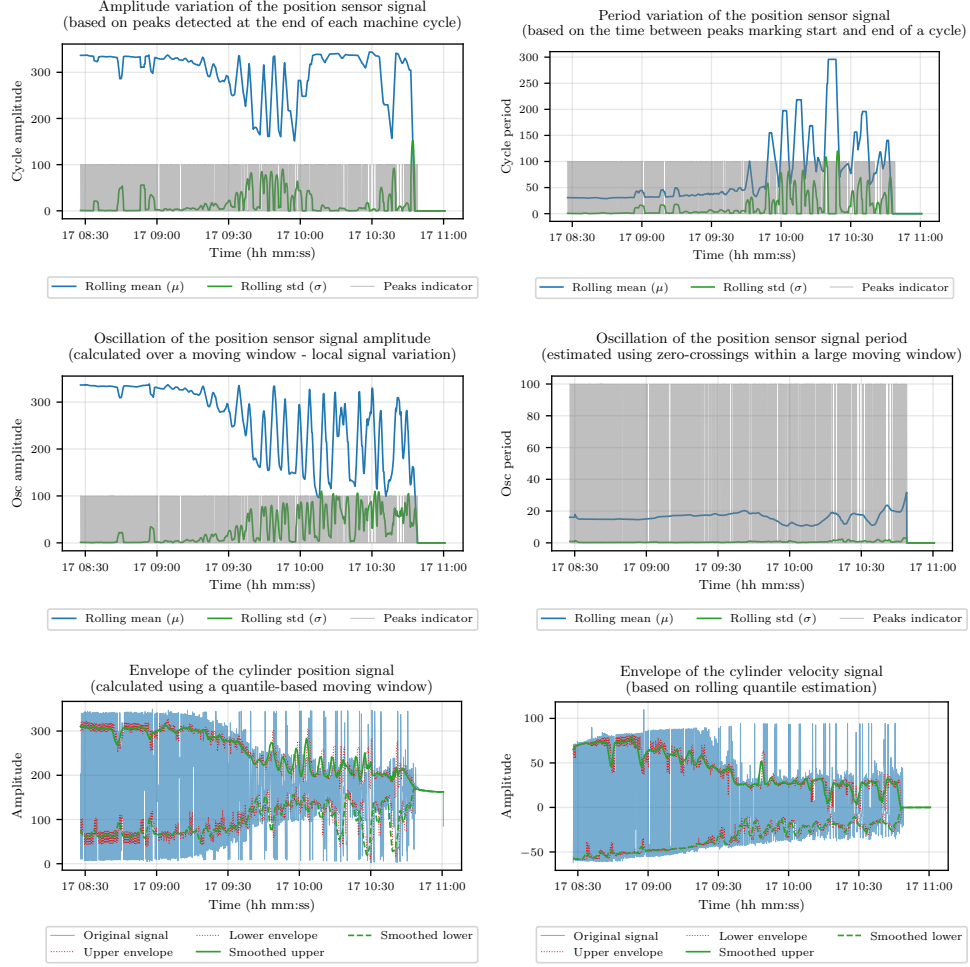


Fig. 5: Extracted features derived from the hydraulic cylinder position and velocity signals. Rolling metrics characterize cycle- and oscillation-level dynamics, while envelope-based features capture amplitude variability over time.

Figure 6 shows an excerpt of three extracted variables over a minute-scale interval. The signals exhibit rapid oscillatory behavior, which is characteristic of hydraulic systems operating under closed-loop control and high-pressure conditions.

The signal denoted as L_{alarm} corresponds to a controlled leakage alarm introduced during the preprocessing stage. This leakage is manually simulated by opening valve Y3 in the hydraulic circuit (see Fig. 1), allowing the evaluation of detection sensitivity under well-defined fault conditions.

In Figures 5, the envelope of the variable was determined using upper quartiles. This approach was adopted to evaluate variability, as it characterizes the amplitude fluctuations of the signal over time.

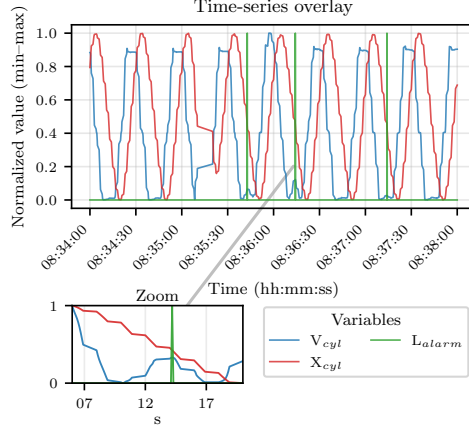


Fig. 6: Excerpt of selected hydraulic time-series signals.

After analyzing the variables, those listed in Table 9 are used for the classification process.

Table 9: Input features used in clustering and state modeling.

Variable Name	Description
osc_amplitude_rolling_mean	Rolling mean of the oscillation amplitude.
vel_cylinder_envelope_upper_smooth	Smoothed upper envelope of the cylinder velocity signal.
vel_cylinder_envelope_lower_smooth	Smoothed lower envelope of the cylinder velocity signal.
osc_period_rolling_mean	Rolling mean of the oscillation period based on zero-crossing detection.
cycle_period_rolling_mean	Estimated period between consecutive peaks representing full cylinder cycles.
pos_clean_envelope_lower_smooth	Smoothed lower envelope of the cleaned cylinder position signal.

Note: All features are computed from cleaned signals and normalized prior to clustering and state modeling.

To assess unsupervised state segmentation from complementary perspectives, two conceptually distinct models are employed. K-Means clustering is used as a baseline to evaluate the *geometric separability* of operational states in the feature space, independent of temporal ordering. In contrast, the Hidden Markov Model (HMM) explicitly models *temporal coherence* and state persistence, which is critical for stable RUL estimation. Together, both methods provide a benchmark comparison between instantaneous geometric clustering and sequence-aware probabilistic state modeling.

5.3.2 Model 1: K-Means for State Clustering

The clustering algorithm employed is presented in Algorithm 6, and the corresponding parameter definitions are provided in Table 10. To determine the optimal number of

clusters for effective data differentiation, the elbow method was applied, as illustrated in Figure 7.

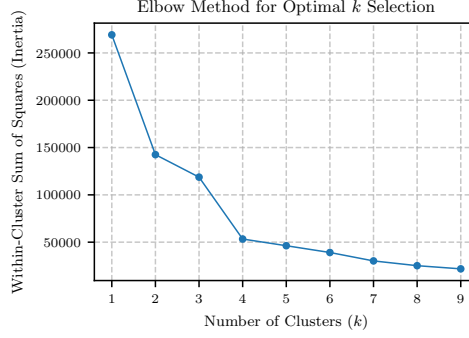


Fig. 7: Elbow method for cluster optimization

Algorithm 6 K-Means Clustering for Signal Segmentation

Require: Feature matrix $\mathbf{X} \in \mathbb{R}^{T \times 6}$

Ensure: Cluster assignment vector $\mathbf{z} \in \{0, \dots, k-1\}^T$

- 1: Normalize $\mathbf{X}$ using standard scaling
- 2: Compute $WCSS(k)$ for $k = 1, \dots, 9$
- 3: Select $k = 4$ using the Elbow criterion
- 4: Solve

$$\min_{\{\mathbf{c}_j\}_{j=1}^k} \sum_{j=1}^k \sum_{\mathbf{x}_t \in \mathcal{C}_j} \|\mathbf{x}_t - \boldsymbol{\mu}_j\|_2^2$$

- 5: Assign each $\mathbf{x}_t$ to its nearest centroid
 - 6: Store trained model
 - 7: **return** $\mathbf{z}$
-

The selected features are low-dimensional, cycle-synchronous descriptors that emphasize distributional variability rather than absolute signal levels. This choice is consistent with the Elbow-based selection of four operating states, as these features maximize inter-cluster separation while preserving compact and interpretable cluster centers. Additional parameters were intentionally excluded to avoid redundant correlations and to ensure stable centroid distributions under varying operating conditions.

Table 10: K-Means clustering configuration.

Parameter	Value / Description
Number of clusters (k)	$k = 4$, selected using the Elbow method based on within-cluster sum of squares.
Initialization method	k-means++ for improved and stable centroid initialization.
Distance metric	Euclidean distance in normalized feature space.
Optimization algorithm	Elkan algorithm, efficient for dense and low-dimensional data.
Random seed	Fixed seed (2025) to ensure reproducibility of clustering results.

Note: All features are standardized prior to clustering, and the trained K-Means model is stored for consistent reuse during inference.

5.3.3 Model 2: HMM for State Classification

For the definition of states in the HMM, four states were employed. Similar to the cluster configuration, this setup validated the differentiation achieved with K-Means and evaluated machine states and operating conditions, with the algorithm and model parameters summarized in Algorithm 7 and Table 11.

Algorithm 7 Hidden Markov Model for State Estimation

Require: Normalized feature matrix $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_T)$, $\mathbf{x}_t \in \mathbb{R}^6$

Ensure: Most probable hidden state sequence $\mathbf{s} = (s_1, \dots, s_T)$

- 1: Initialize a Gaussian HMM with $n = 4$ hidden states
- 2: Define state transition matrix $\mathbf{A}$ and initial distribution $\boldsymbol{\pi}$
- 3: Specify Gaussian emission densities $p(\mathbf{x}_t | s_t = i) = \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$ with diagonal covariance $\boldsymbol{\Sigma}_i$
- 4: Estimate model parameters $\{\mathbf{A}, \boldsymbol{\pi}, \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i\}$ by maximizing the log-likelihood

$$\log p(\mathbf{X} | \lambda) = \sum_{t=1}^T \log \left(\sum_{s_t} p(\mathbf{x}_t | s_t) p(s_t | s_{t-1}) \right)$$

using the Baum–Welch algorithm (maximum 1000 iterations)

- 5: Compute the most probable hidden state sequence $\mathbf{s}^* = \arg \max_{\mathbf{s}} p(\mathbf{s} | \mathbf{X})$ using the Viterbi algorithm
 - 6: Store trained HMM model
 - 7: **return** $\mathbf{s}^*$
-

Table 11: Hidden Markov Model configuration.

Parameter	Value / Description
Number of hidden states (n)	$n = 4$, representing discrete operational regimes inferred from the feature space.
Observation model	Multivariate Gaussian emission densities $p(\mathbf{x}_t s_t = i) = \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$.
Covariance structure	Diagonal covariance matrices $\boldsymbol{\Sigma}_i$, assuming conditional independence between features.
Training algorithm	Baum-Welch algorithm for maximum-likelihood parameter estimation.
Maximum iterations	Limited to 1000 iterations to ensure convergence of the log-likelihood.
Decoding algorithm	Viterbi algorithm for most probable hidden state sequence inference.
Random seed	Fixed seed (2025) to ensure reproducibility of training and decoding.

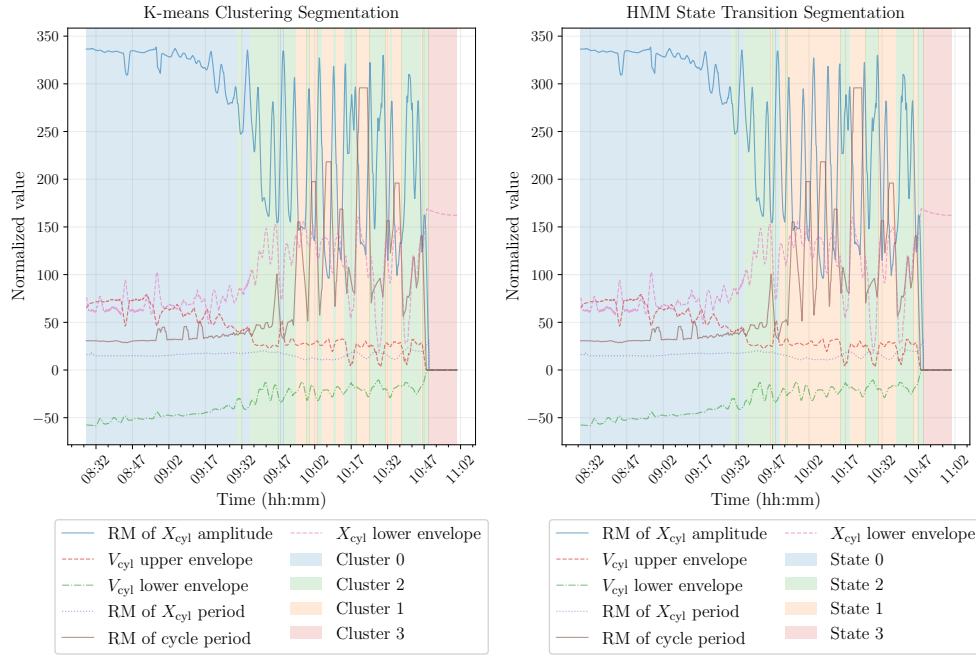
Note: The HMM operates on normalized features and estimates discrete system states by probabilistic sequence modeling.

5.3.4 Results Classification: HMM & Kmeans

Detecting anomalous machine conditions requires a clear differentiation of operational states. Traditionally, this relies on expert knowledge; however, unsupervised machine learning methods such as K-Means and Hidden Markov Models (HMM) enable automated clustering of behavior patterns based on derived features [50].

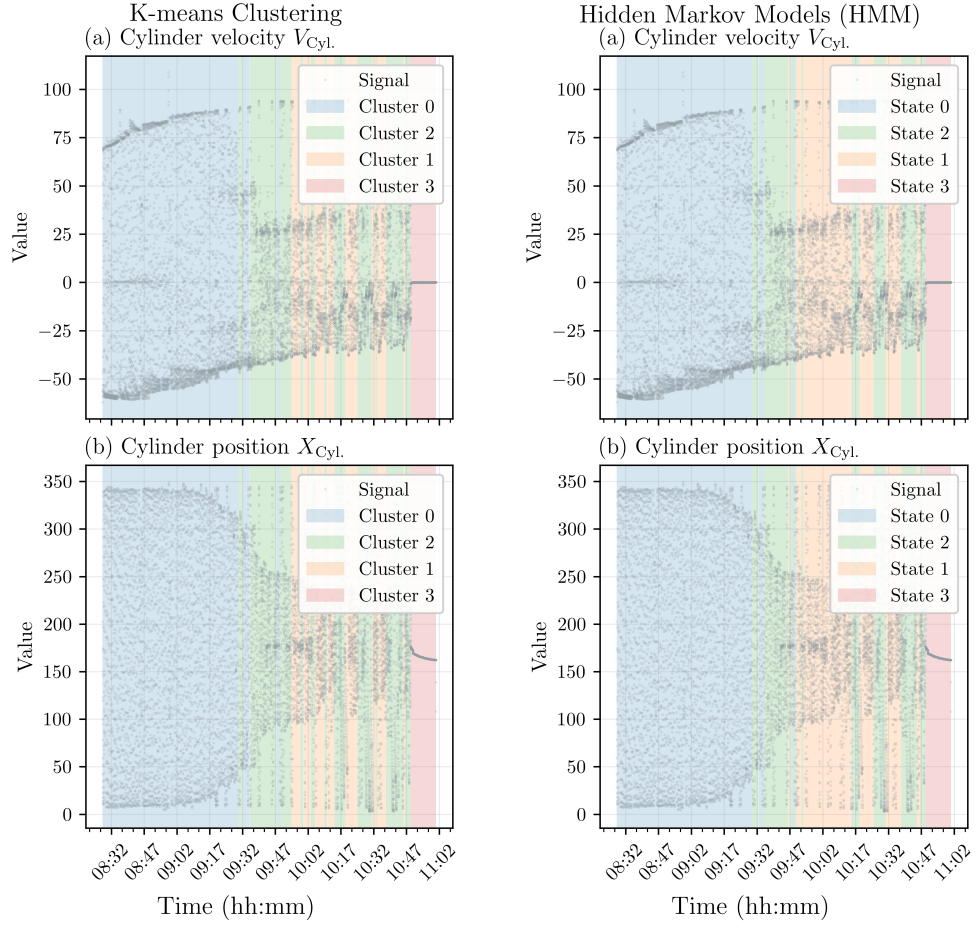
Figure 8a illustrates state segmentation using K-Means on envelope and velocity features, revealing clusters without prior labeling. Conversely, Figure 8b demonstrates the HMM application, which incorporates sequential dependencies and probabilistic transitions. Their comparison in Figures 8 highlights that K-Means yields sharper cluster boundaries, whereas HMM provides smoother state transition modeling.

The analysis relies on derived features, where $\text{RM}(\cdot)$ denotes a rolling mean used to smooth short-term fluctuations, and upper/lower envelopes capture the bounding trajectories of X_{cyl} and V_{cyl} .



(a) State segmentation using K-Means cluster(b) State segmentation using Hidden Markov ing on derived features. Models (HMM) on the same features.

Fig. 8: Comparison of unsupervised state segmentation approaches applied to hydraulic cylinder signals.



(a) K-means-based cluster assignment for machine state segmentation. (b) Hidden Markov Model-based state sequence inferred from clustered features.

Fig. 9: Comparison of unsupervised state segmentation methods applied to hydraulic cylinder signals.

For the failure modes *Excessive Internal Leakage* and *Anomalous Operation (Slow-down)*, latent operating states emerging from unsupervised clustering were interpreted and refined using expert knowledge (Table 12). These states are not supervised labels, but semantic representations of cluster-wise statistical patterns grounded in established hydraulic actuator behavior. Based on this definition, the first RUL corresponds to the transition from *Normal Operation* to *Abnormal Operation*, and the second (critical) RUL to the transition from *Abnormal Operation* to *Critical Operation*, both identified via HMM due to its stability and reduced temporal fluctuations. Although a linear degradation assumption is adopted, the pipeline supports periodic supervised retraining to accommodate nonlinear or abrupt changes as new data become available.

Table 12: Description of states/clusters used in modeling.

State / Cluster	Description
0	Normal operation.
1	Critical operation.
2	Abnormal operation.
3	Standstill.

5.4 Remaining Useful Life Estimation

The input features used for RUL prediction are summarized in Table 13. Window-based velocity descriptors (e.g., `Vel_cy_max` and `Vel_cy_min`) are computed using only the last 30 samples prior to the current time instant (≈ 3 s at 100 ms sampling), providing short-term dynamic context while remaining negligible compared to the minute-level RUL horizon; therefore, no future information is introduced.

Table 13: Input features used for RUL prediction.

Variable name	Description
<code>CtrlX_2_Q</code>	Oil flow rate (Q), derived from the hydraulic circuit measurements.
<code>Pa_Ps_diff</code>	Pressure ratio between port A and supply pressure (P_A/P_S), indicative of load and leakage conditions.
<code>Pb_Pa_diff</code>	Pressure ratio between port B and port A (P_B/P_A), sensitive to internal leakage and abnormal pressure regimes.
<code>vel_smoothed</code>	Smoothed retraction velocity of the cylinder, emphasizing degradation-related slowdowns.
<code>vel_cylinder</code>	Instantaneous cylinder velocity measured by the position sensor derivative.
<code>pos_clean</code>	Cleaned cylinder position signal after interpolation and outlier removal.
<code>Vel_cy_max</code>	Maximum cylinder velocity computed within a 30-step future window, capturing transient performance peaks.
<code>Vel_cy_min</code>	Minimum cylinder velocity computed within a 30-step future window, reflecting worst-case dynamic behavior.

RUL estimation is conducted through a state-based classification strategy. This approach predicts the remaining time until a state transition occurs, focusing particularly on forecasting the time required to reach an undesired operating condition, as defined in Section 5.3.4. The RUL formulation illustrated in Figure 10 has been adopted due to its strong alignment with real-world industrial scenarios, enabling both scalable deployment and generalized applicability. The input variables used for RUL prediction, listed in Table 13, are readily available from the sensor infrastructure and can be continuously fed into the system to support real-time prognostics.

RUL as a State-Transition Problem

Classical RUL often extrapolates time-to-failure, but industrial hydraulic assets are rarely operated until catastrophic breakdown. Degradation is typically observed as progressive performance drift (e.g., leakage-driven velocity loss or pressure imbalance), and maintenance is triggered when operation becomes undesired or uneconomical. Therefore, RUL is formulated here as the remaining time until a predefined *undesired operating state* is reached, aligning prognostics with operational decision-making rather than terminal failure prediction.

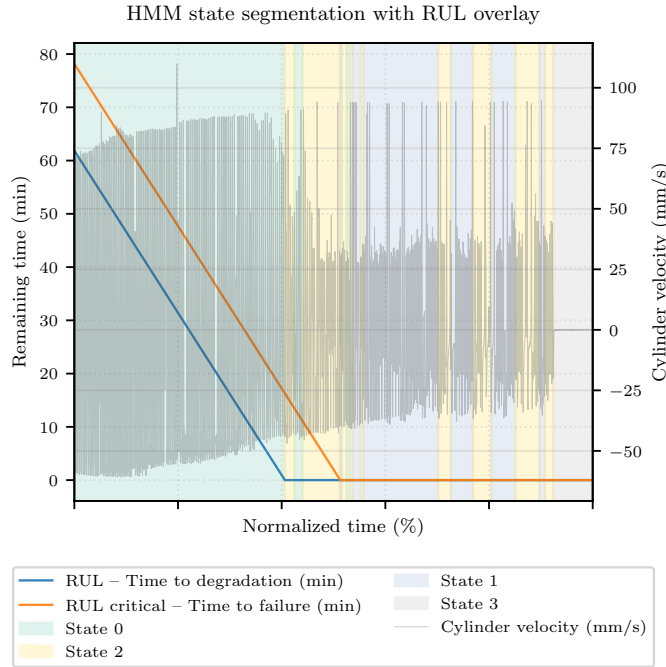


Fig. 10: RUL estimation based on state transitions.

5.4.1 Model 3: XGBoost Regressor for RUL Prediction

An extensive hyperparameter optimization was conducted to determine the best-performing configuration, as detailed in Table 14. This configuration was subsequently employed for the prediction of both *RUL* and *RUL_critical*. The complete training algorithm is presented in Algorithm 8.

Table 14: XGBoost configuration for RUL prediction.

Parameter	Value
Number of estimators	200 boosting rounds
Learning rate	0.05
Early stopping rounds	20 (validation set)
Parallel jobs	All available cores ($n_{\text{jobs}} = -1$)
Evaluation metric	Validation MAE
Random seed	42
Model type	XGBRegressor

Algorithm 8 XGBoost Training Pipeline for RUL Prediction

Require: Normalized dataset $\mathbf{D}$ with feature matrix $\mathbf{X}$ and targets $\mathbf{y}_1$ (RUL) and $\mathbf{y}_2$ (critical RUL)

Ensure: Trained models $\hat{f}_{\text{RUL}}$, $\hat{f}_{\text{RUL}_c}$, evaluation metrics, and SHAP/PDP visualizations

- 1: **Data preprocessing**
- 2: Define input features $\mathbf{X} = \{x_1, \dots, x_6\}$ and targets $\mathbf{y}_1, \mathbf{y}_2$
- 3: Remove missing values in $\mathbf{X}$, $\mathbf{y}_1$, $\mathbf{y}_2$, and timestamps
- 4: Normalize features using standard scaling $\tilde{\mathbf{X}} = (\mathbf{X} - \boldsymbol{\mu})/\boldsymbol{\sigma}$
- 5: Store the fitted scaler for reproducibility
- 6: **Train/validation/test split**
- 7: Concatenate $\tilde{\mathbf{X}}$, $\mathbf{y}_1$, $\mathbf{y}_2$, and timestamps into dataset $\mathbf{Z}$
- 8: Split $\mathbf{Z}$ into training (70%), validation (10%), and test (20%) subsets
- 9: Extract input matrices and target vectors for each subset
- 10: **Model training**
- 11: Train XGBoost model $\hat{f}_{\text{RUL}}$ on $(\tilde{\mathbf{X}}_{\text{train}}, \mathbf{y}_{1,\text{train}})$ using early stopping
- 12: Train XGBoost model $\hat{f}_{\text{RUL}_c}$ on $(\tilde{\mathbf{X}}_{\text{train}}, \mathbf{y}_{2,\text{train}})$
- 13: Store trained models to disk
- 14: **Model evaluation**
- 15: **for** $\mathcal{S} \in \{\text{train, val, test}\}$ **do**
- 16: Predict $\hat{\mathbf{y}}$ using $\hat{f}$ on subset $\mathcal{S}$
- 17: Compute MAE, RMSE, and coefficient of determination R^2
- 18: Compute NASA PHM score with asymmetric penalties (α, β)
- 19: Store prediction plots with timestamps $\mathbf{t}$ and true targets $\mathbf{y}$
- 20: **end for**
- 21: **Explainability analysis**
- 22: Compute SHAP values for $\tilde{\mathbf{X}}$
- 23: Generate SHAP dependence plots and partial dependence plots (PDPs)
- 24: Store all explainability visualizations
- 25: **return** $\hat{f}_{\text{RUL}}$, $\hat{f}_{\text{RUL}_c}$, evaluation metrics, SHAP and PDP results

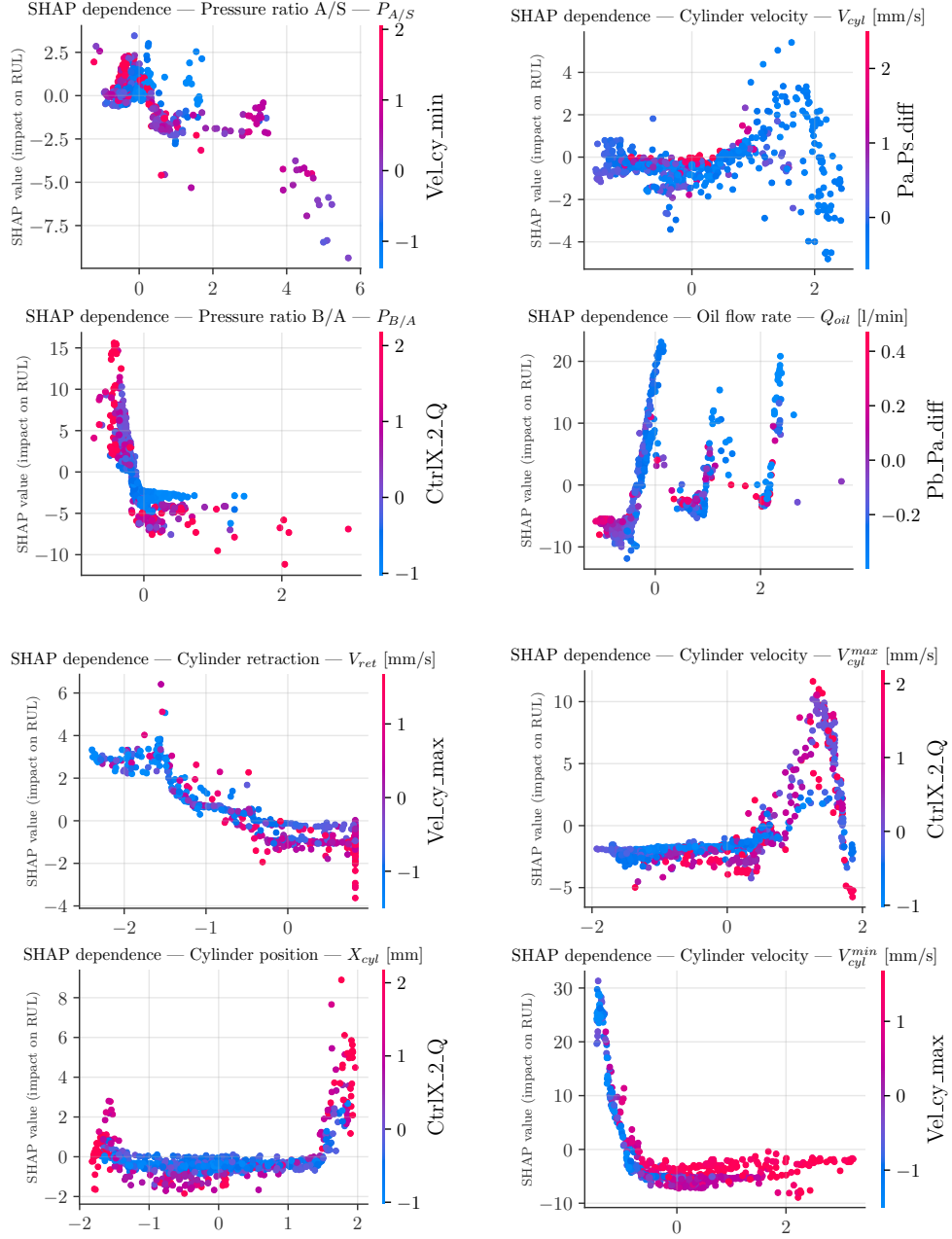


Fig. 11: SHAP dependence plots showing the local influence of the most relevant features on RUL prediction using XGBoost.

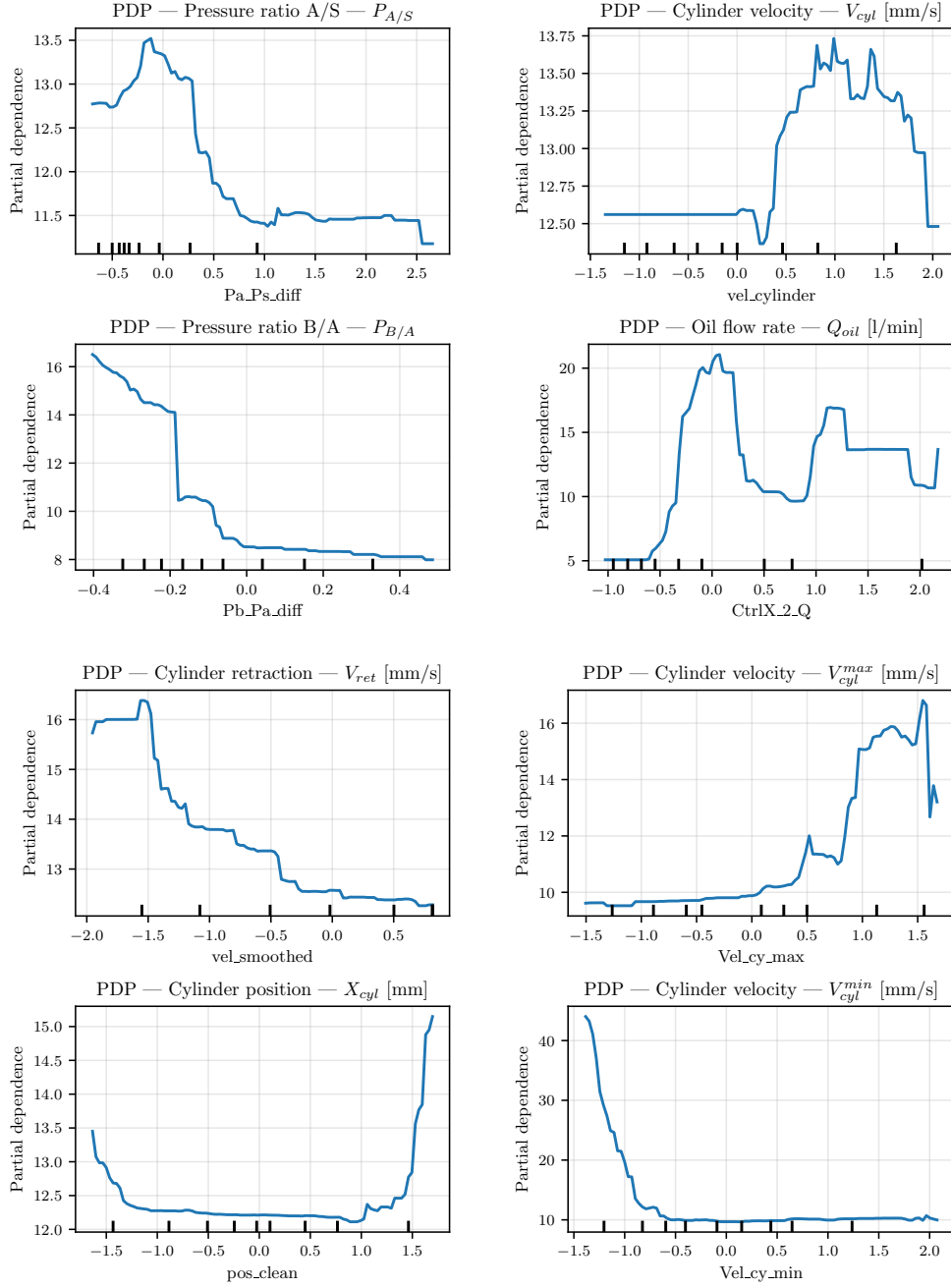


Fig. 12: Partial Dependence Plots (PDP) illustrating the marginal effect of the most influential features on RUL prediction using XGBoost.

5.4.2 Results Model 3: XGBoost Regressor

Table 15 summarizes the model performance, including a PHM-based metric adapted for minute-level RUL evaluation of the hydraulic system, computed following Algorithm 9 and the formulation in [51].

Algorithm 9 PHM score computation.

Require: Ground truth $\mathbf{y}_{\text{true}} = (y_{\text{true},1}, \dots, y_{\text{true},N})$, predictions $\mathbf{y}_{\text{pred}} = (y_{\text{pred},1}, \dots, y_{\text{pred},N})$

Require: Parameters Δ_{crit}^+ , Δ_{tol}^- , penalty levels P_+ , P_-

Ensure: PHM_{tot} and PHM_{avg}

1: Compute penalty scales

$$\beta \leftarrow \frac{\Delta_{\text{crit}}^+}{\ln(1 + P_+)}, \quad \alpha \leftarrow \frac{\Delta_{\text{tol}}^-}{\ln(1 + P_-)}.$$

2: Compute residuals for $i = 1, \dots, N$

$$d_i \leftarrow y_{\text{pred},i} - y_{\text{true},i}.$$

3: Compute asymmetric cost for each residual

$$s_i \leftarrow \begin{cases} \exp(-d_i/\alpha) - 1, & d_i < 0 \quad (\text{conservative}) \\ \exp(d_i/\beta) - 1, & d_i \geq 0 \quad (\text{optimistic}). \end{cases}$$

4: Aggregate PHM scores

$$\text{PHM}_{\text{tot}} \leftarrow \sum_{i=1}^N s_i, \quad \text{PHM}_{\text{avg}} \leftarrow \frac{\text{PHM}_{\text{tot}}}{N}.$$

Note: Lower values indicate better performance. Optimistic errors ($d_i \geq 0$) incur higher penalties than conservative errors ($d_i < 0$). PHM denotes **Prognostics and Health Management**.

5: **return** PHM_{tot} , PHM_{avg}

Table 15: Performance metrics for RUL prediction using XGBoost models.

Model	Target	Set	MAE	RMSE	R^2	PHM_{tot}	PHM_{avg}
XGB	RUL	Train	1.081	2.412	0.984	737.215	0.024
XGB	RUL	Val	1.289	3.090	0.973	128.842	0.030
XGB	RUL	Test	1.297	3.043	0.975	258.345	0.030
XGB	RUL_c	Train	1.537	3.048	0.986	1056.575	0.035
XGB	RUL_c	Val	1.805	3.715	0.978	181.312	0.042
XGB	RUL_c	Test	1.794	3.709	0.979	358.718	0.041

Note: XGB denotes XGBoost. RUL_c refers to critical RUL. PHM scores follow the NASA asymmetric penalty formulation, penalizing optimistic errors more strongly than conservative ones.

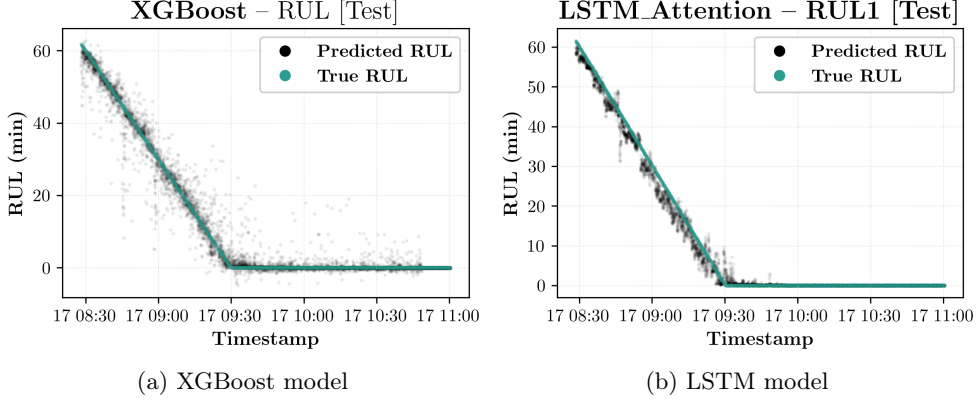


Fig. 13: Comparison of predicted versus ground-truth RUL values on the test set using two learning paradigms: (a) gradient boosting (**XGBoost**) and (b) deep recurrent learning (**LSTM**). Both approaches demonstrate strong generalization to unseen data, with XGBoost capturing abrupt degradation transitions and LSTM providing smoother long-term trends.

Feature Explainability and Selection

Feature explainability was addressed using SHAP values [52] to quantify local, sample-level impact, and Partial Dependence Plots (PDPs) [53] to analyze global marginal effects. These complementary methods highlight the most influential features from both local and global perspectives.

Figures 11 present SHAP dependence plots, and Figure 14 shows the aggregated SHAP summary. Partial dependence plots (PDPs) for the same features are provided in Figures 12, illustrating the average model response to independent feature variations.

All interpretability results correspond to the model trained for *RUL* prediction. Equivalent plots for *RUL_critical* are omitted due to analogous behavior, but performance metrics are reported in Table 15.

The most influential features—oil flow rate, minimum and maximum cylinder velocities, and pressure ratio (P_B/P_A)—reflect the expected physical behavior under hydraulic leakage and rising temperature, where pressure drops induce chamber imbalances affecting both velocity and displacement. Figure 13 show predicted versus ground-truth RUL for the test set.

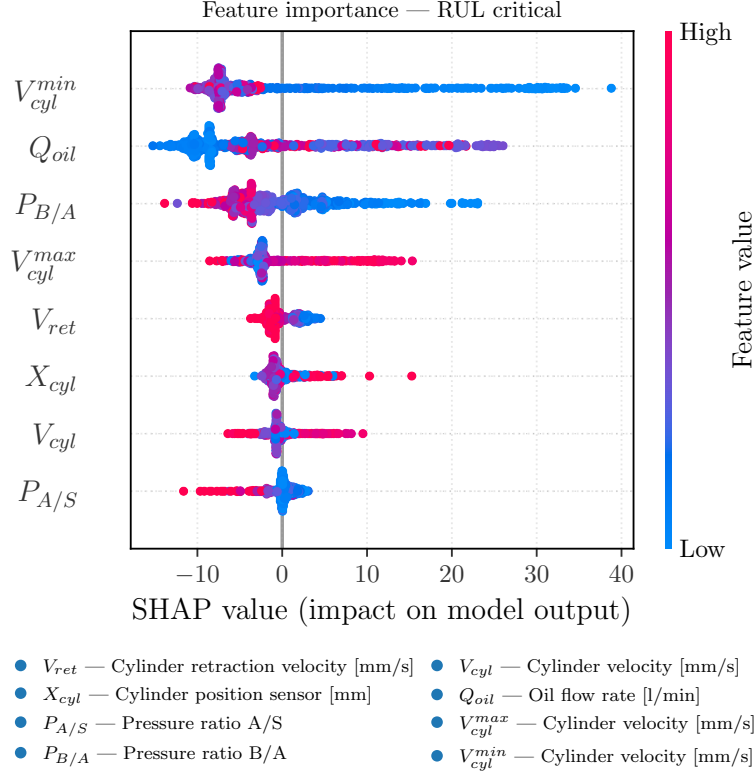


Fig. 14: SHAP summary plot showing global feature importance for RUL prediction using XGBoost.

5.4.3 Model 4: LSTM-Attention for RUL Estimation

The choice of a LSTM network with an attention mechanism follows the recommendation in [54], which highlights its suitability for regression tasks where past events strongly influence the current condition.

The detailed parameter configuration of the LSTM model is provided in Table 16. The mathematical formulation of the attention mechanism is presented in Algorithm 11, while the complete training and evaluation pipeline is outlined in Algorithm 11.

Algorithm 10 Multi-Head Self-Attention

Require: Input tensor $\mathbf{X} \in \mathbb{R}^{B \times T \times F}$

Ensure: Context-aware representation $\mathbf{Z} \in \mathbb{R}^{B \times T \times d_{\text{model}}}$

1: **Linear projections**

2: $\mathbf{Q} = \mathbf{XW}^Q \quad \mathbf{K} = \mathbf{XW}^K \quad \mathbf{V} = \mathbf{XW}^V$

3: Reshape into H heads with dimensions d_k, d_v

4: **Scaled dot-product attention**

$$\mathbf{O}_h = \text{softmax} \left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_k}} \right) \mathbf{V}_h$$

5: **Concatenate heads**

$$\mathbf{O} = \text{Concat}(\mathbf{O}_1, \dots, \mathbf{O}_H)$$

6: **Output projection**

$$\mathbf{Z} = \mathbf{O}\mathbf{W}^O$$

7: **return** $\mathbf{Z}$

Table 16: LSTM + Multi-Head Attention configuration for RUL prediction.

Parameter	Value
Model type	Keras Functional API: LSTM sequence encoder with multi-head self-attention and dense regression head.
Input shape	(W, F) , where $W = \text{WINDOW_SIZE}$ and $F = \mathbf{X}_{\text{cols}} $.
LSTM layer	LSTM with 128 units, <code>return_sequences=True</code> , activation <code>tanh</code> .
Normalization (after LSTM)	<code>LayerNormalization</code> .
Dropout (after LSTM)	0.2.
Attention mechanism	<code>MultiHeadAttention</code> with $H = 4$ heads, key dimension $d_k = 64$, applied as self-attention on the sequence.
Normalization (after attention)	<code>LayerNormalization</code> .
Dropout (after attention)	0.2.
Pooling layer	<code>GlobalAveragePooling1D</code> for sequence-level embedding.
Dense head	<code>Dense(64, ReLU) → Dropout(0.1) → Dense(32, ReLU)</code> .
Output layer	<code>Dense(1)</code> with <code>softplus</code> activation to enforce non-negative RUL predictions.
Optimizer	Adam optimizer with learning rate 1×10^{-3} .
Loss function	Mean Squared Error (MSE).
Evaluation metric	Mean Absolute Error (MAE).
Training epochs	31 epochs.
Batch size	64.
Callbacks	<code>EarlyStopping</code> (patience = 10, monitor <code>val_loss</code> , restore best weights); <code>ReduceLROnPlateau</code> (factor = 0.1, patience = 5, monitor <code>val_loss</code>); <code>EpochPrinter</code> (log every 200 batches).
Model persistence	Trained model saved as a <code>.keras</code> file for each target.

Algorithm 11 LSTM + Multi-Head Self-Attention Training and Evaluation Pipeline

Require: Dataset $\mathbf{D}$ with features $\mathbf{X}$, targets $\mathbf{y}_1 = \text{RUL}_{\min}$, $\mathbf{y}_2 = \text{RUL}_{\text{crit}, \min}$, timestamps $\mathbf{t}$
Ensure: Trained models $\hat{g}_{\text{RUL1}}$, $\hat{g}_{\text{RULcrit}}$, fitted scaler, metrics, and figures

- 1: **Preprocess and split**
- 2: Remove rows containing NaNs in $(\mathbf{X}, \mathbf{y}_1, \mathbf{y}_2, \mathbf{t})$
- 3: Split into train/validation/test using `test_size` and `val_size` (validation computed relative to the non-test subset)
- 4: Fit `StandardScaler` on $\mathbf{X}_{\text{train}}$ and transform all splits; store the fitted scaler
- 5: **Sequence windowing**
- 6: Given window length W , construct sequences $(\mathbf{S}, \mathbf{y}, \mathbf{t})$ by pairing each label and timestamp with the last step of its window
- 7: **Model definition (LSTM + multi-head attention)**
- 8: Build model $\mathcal{M}$ according to Table 16:
 - Input in $\mathbb{R}^{W \times F}$ processed by an LSTM with 128 units
 - Apply `MultiHeadAttention` with $H = 4$ heads and $d_k = 64$ per head
 - Layer normalization, dropout, `GlobalAveragePooling1D`, and two dense layers
 - Output $\hat{y} \in \mathbb{R}_{\geq 0}$ via `softplus` activation
 - Train with Adam optimizer, MSE loss, MAE metric, early stopping, and LR scheduler
- 9: **Training for both targets**
- 10: **for** $k \in \{1, 2\}$ **do**
- 11: Train model $\hat{g}_k$ on $(\mathbf{S}_{\text{train}}, \mathbf{y}_{k, \text{train}})$ and validate on $(\mathbf{S}_{\text{val}}, \mathbf{y}_{k, \text{val}})$ using the hyperparameters and callbacks in Table 16
- 12: Store model weights (`.keras`), training history (JSON), and loss/MAE plots
- 13: **end for**
- 14: **Evaluation and visualization**
- 15: **for** $k \in \{1, 2\}$ **do**
- 16: **for** $\mathcal{S} \in \{\text{Train, Validation, Test}\}$ **do**
- 17: Predict $\hat{\mathbf{y}}$ for split $\mathcal{S}$ and clip to $\hat{\mathbf{y}} \geq 0$
- 18: Sort predictions by timestamps to align $(\mathbf{t}, \mathbf{y}_k, \hat{\mathbf{y}})$
- 19: Compute MAE, RMSE, and coefficient of determination R^2
- 20: Compute NASA PHM score $(\text{PHM}_{\text{tot}}, \text{PHM}_{\text{avg}})$ using (α, β) from `pick_alpha_beta_minutes`
- 21: Store timestamped prediction plots
- 22: **end for**
- 23: **end for**
- 24: **return** $\hat{g}_{\text{RUL1}}$, $\hat{g}_{\text{RULcrit}}$, scaler, metrics, figures

5.4.4 Results model 4: LSTM-Attention

To evaluate the impact of the attention mechanism on convergence and model performance, we conducted 25 independent training runs (10 epochs each) using the architecture detailed in Table 16. For comparison, the attention layer was replaced only in the evaluation scenario by a global average pooling operation ($x = \text{tf.keras.layers.GlobalAveragePooling1D}()(x)$), while keeping all other hyperparameters and training settings identical. The averaged results are reported in Table 17.

Table 17: Impact of the attention mechanism on LSTM-based RUL prediction (average over 25 runs, 10 epochs per run).

Model	Target	Set	MAE	RMSE	R^2	PHM _{tot}	PHM _{avg}
LSTM	RUL	Train	6.518	11.063	0.661	5086.928	0.167
LSTM	RUL	Val	6.361	10.830	0.667	702.021	0.162
LSTM	RUL	Test	6.576	11.212	0.660	1465.520	0.169
LSTM+Attn	RUL	Train	2.468	5.554	0.915	1785.685	0.059
LSTM+Attn	RUL	Val	2.502	5.681	0.908	258.529	0.060
LSTM+Attn	RUL	Test	2.558	5.761	0.910	533.447	0.061
LSTM	RUL _c	Train	7.690	12.685	0.750	6324.223	0.208
LSTM	RUL _c	Val	7.548	12.364	0.758	873.121	0.202
LSTM	RUL _c	Test	7.749	12.874	0.747	1825.361	0.210
LSTM+Attn	RUL _c	Train	3.882	7.539	0.912	2877.344	0.094
LSTM+Attn	RUL _c	Val	3.863	7.541	0.910	408.075	0.094
LSTM+Attn	RUL _c	Test	3.916	7.649	0.911	835.622	0.096

Note: LSTM+Attn denotes an LSTM model with additive attention.

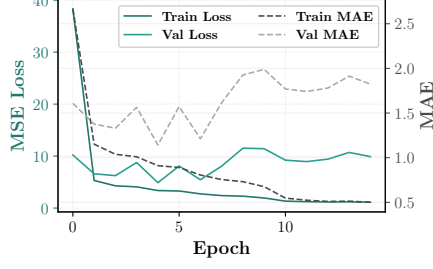
Final performance metrics after 31 training epochs with attention are presented in Table 18. A comprehensive visualization of the model’s behavior is provided: predicted vs. ground-truth RUL in test set (Figs. 13), training history for loss and MAE (Fig. 15), global feature importance (Fig. 17), spatio-temporal attention heatmap (Fig. 16).

In Fig. 16, a consistent attention pattern is visualized, obtained by extracting the weights from the MultiHeadAttention layer. The function builds a submodel to retrieve attention scores, averages them across heads and samples, and computes the per-timestep attention vector used in the heatmap.

Table 18: Performance metrics for RUL prediction using LSTM with attention mechanism (31 training epochs).

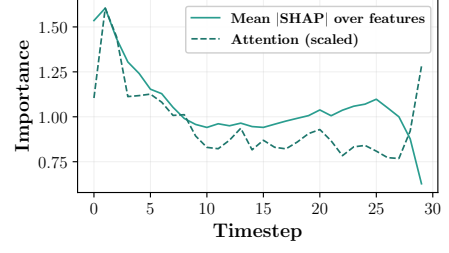
Model	Target	Set	MAE	RMSE	R^2	PHM _{tot}	PHM _{avg}
LSTM+Attn	RUL	Train	0.811	1.810	0.991	505.632	0.017
LSTM+Attn	RUL	Val	1.142	2.210	0.986	104.005	0.024
LSTM+Attn	RUL	Test	0.977	1.804	0.991	172.400	0.020
LSTM+Attn	RUL _c	Train	1.027	1.770	0.995	630.074	0.021
LSTM+Attn	RUL _c	Val	1.825	3.104	0.985	168.070	0.039
LSTM+Attn	RUL _c	Test	1.415	2.420	0.991	251.242	0.029

RUL1_Training_History – Loss & MAE



(a) Training loss and MAE evolution

Temporal importance — RUL1_test



(b) Temporal SHAP vs. attention weights

Fig. 15: (a) Training loss progression across 31 epochs for the LSTM + Attention model, showing convergence and stability of both loss and MAE metrics. (b) Comparison between temporal SHAP importance values and normalized multi-head attention weights for RUL₁ prediction on the test set, revealing consistent temporal focus between explainability and attention mechanisms.

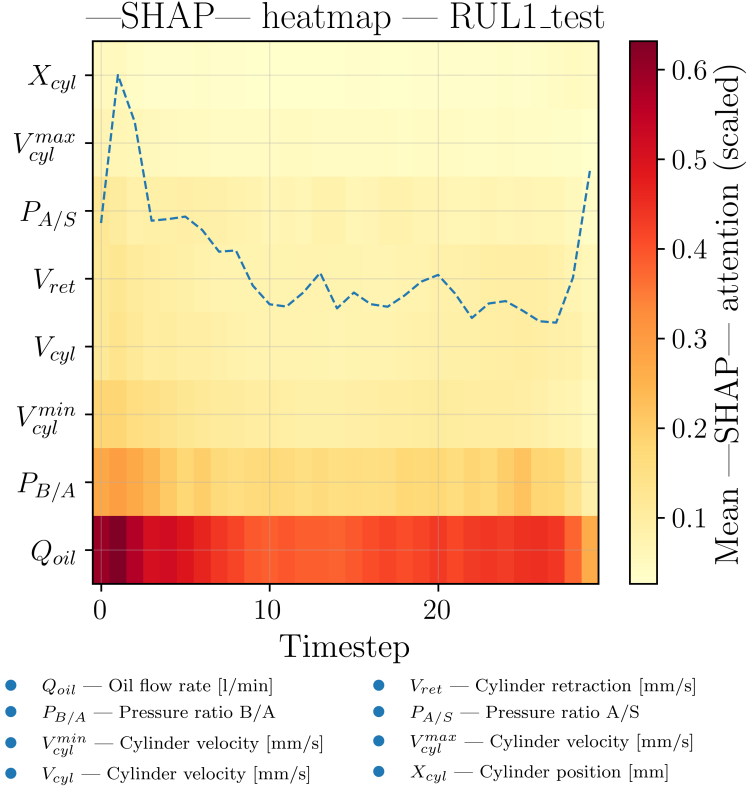


Fig. 16: Spatio-temporal attention heatmap across features and timesteps. High attention weights indicate stronger contribution to the final prediction.

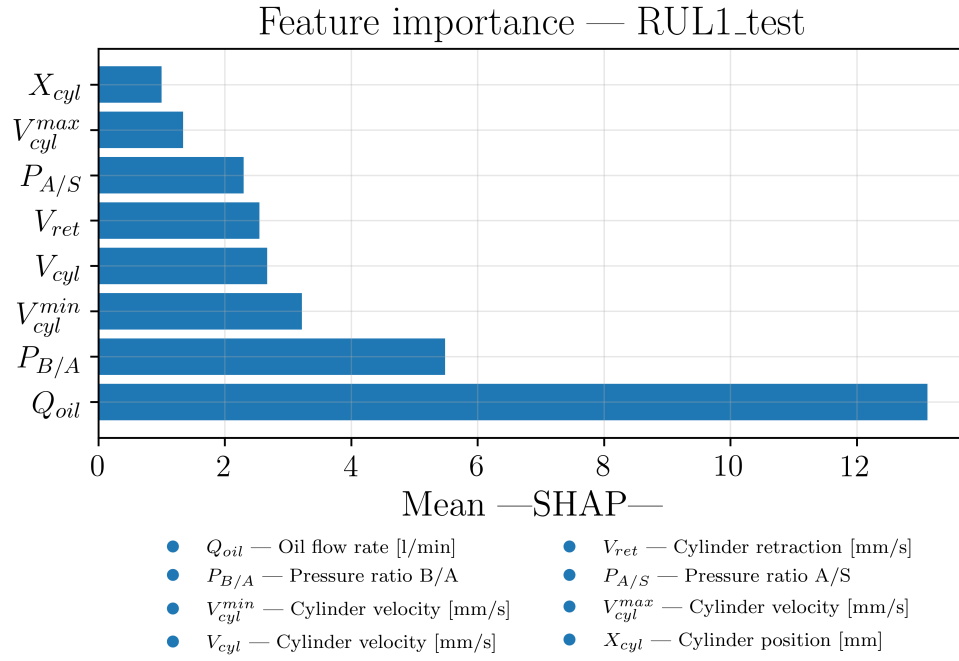


Fig. 17: SHAP-based global feature importance on the test set for the LSTM + Attention model.

6 Limitations and Scope

The proposed framework is validated on a real electro-hydraulic system operated in a controlled commissioning environment. While the hardware, PLC logic, and data acquisition stack are fully industrial, the system has not yet been exposed to long-term operation in a continuous production line.

Consequently, explicitly labeled failure events are limited, reflecting common industrial practice where catastrophic failures are rare and actively avoided. RUL is therefore formulated in terms of transitions toward economically non-viable operating states rather than run-to-failure trajectories.

At this stage, the database primarily captures operational stoppages and their associated causes. Full industrial deployment requires the automated integration of maintenance actions and component-level fault records, enabling tighter coupling between sensor data, physical behavior, and maintenance decisions.

The scope of this work is to establish the architectural and data prerequisites for explainable and economically actionable prognostics; model complexity is treated as secondary to data consistency, physical interpretability, and industrial relevance.

7 Discussion

This work demonstrates that predictive maintenance models for hydraulic systems are fundamentally constrained by system observability. Due to aggregated sensing and tightly coupled dynamics, degradation cannot be reliably attributed at the component level without additional instrumentation; system-level modeling therefore represents the only physically consistent abstraction under realistic industrial conditions. The explainability analysis further reveals that interpretability is strictly data-dependent: explanatory methods can only recover physical relationships that are sufficiently encoded in the dataset, and incomplete sensing may yield misleading interpretations. These findings motivate hybrid approaches that integrate structural priors, such as graph-based models, to complement data-driven learning. Standardized semantic representations, e.g., via Asset Administration Shells, emerge as a prerequisite for scalable and transferable prognostics across heterogeneous hydraulic systems.

8 Conclusion

This study proposed a scalable Edge-AI framework for fault diagnosis and RUL prediction in PLC-integrated hydraulic systems, addressing key challenges such as data heterogeneity, limited failure labels, and high-frequency acquisition constraints. Here, *edge-oriented* does not imply hard real-time inference, but architectural proximity to the PLC, enabling autonomous operation, low-latency data access, and deployment without mandatory cloud dependencies.

The main contributions and findings are summarized as follows:

- **Edge-centric data acquisition:** A modular data architecture enables high-frequency (100 ms) sensor collection via OPC UA, capturing transient phenomena

critical for early degradation detection. Containerized microservices ensure seamless deployment, scalability, and integration into existing infrastructures.

- **Hybrid operational state modeling:** Combining K-Means clustering and HMMs allows identification of machine states without historical failure data. Expert-guided annotations improve physical interpretability by linking data-driven clusters to degradation mechanisms such as internal leakage and thermal anomalies.
- **High-performance RUL prediction:** Experimental validation on a Bosch Rexroth ctrlX CORE hydraulic system demonstrated consistent predictive performance across operating regimes ($\text{MAE} < 2$ minutes, $R^2 > 0.96$), indicating that the proposed state-based RUL formulation is numerically stable and coherent with observed operational events.
- **Explainability through attention:** Attention heatmaps revealed the temporal relevance of sensor inputs, providing interpretable insights that enhance trust in decision-making for predictive maintenance. Importantly, the attention-based explanations consistently highlighted variables known to be physically relevant in hydraulic degradation, such as pressure ratios, flow rate, and velocity, supporting expert validation rather than post-hoc interpretation alone.
- **Event-driven operational context:** Structured downtime and operational event logging provided the contextual layer required to interpret data-driven state transitions and validate prognostic outputs.

By redefining RUL as a state-transition problem rather than relying on catastrophic failure data, the framework broadens the applicability of predictive maintenance in real-world industrial settings. Future work will focus on integrating digital twin simulations, extending multi-asset analytics, and evaluating Transformer-based architectures to improve generalization and explainability.

Appendix A Experimental Testbed: Hydraulic Unit

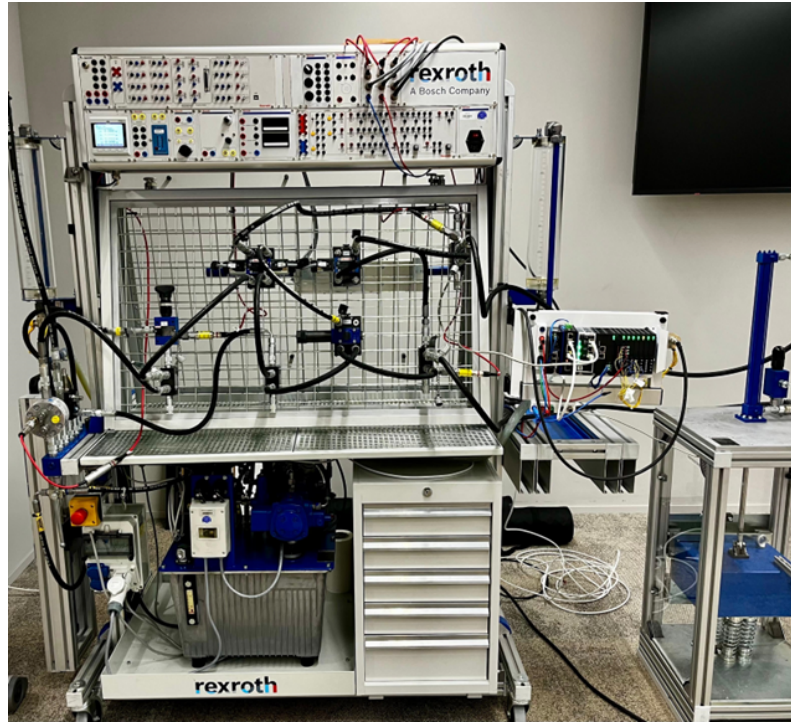


Fig. A1: Schematic of the hydraulic unit used for data acquisition and model evaluation.[48]

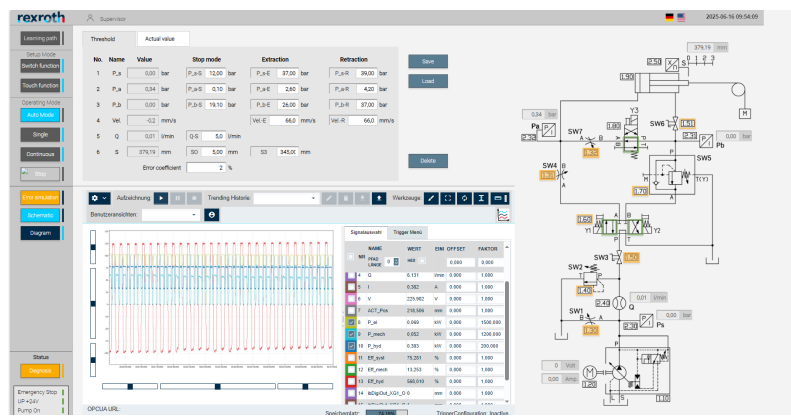


Fig. A2: Operator GUI for visualizing and validating PLC signal activity.[48]

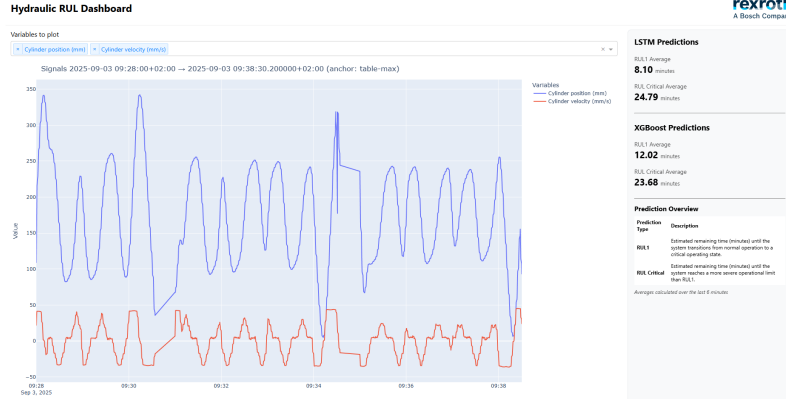


Fig. A3: Visualization of model prediction results for continuous evaluation on test data.

Appendix B Data Flow - Orchestration

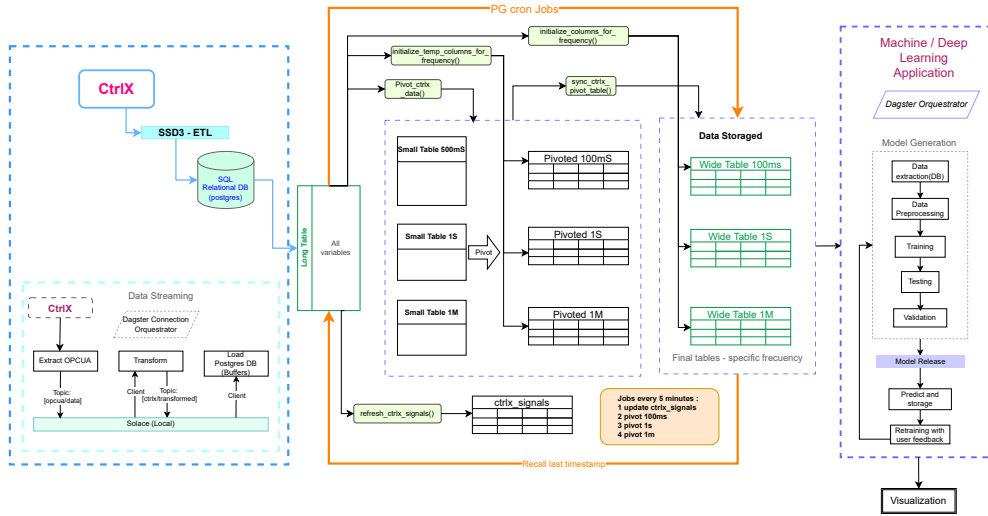


Fig. B4: ETL data flow from PLCs to time-series storage for real-time sensor monitoring and predictive analysis.

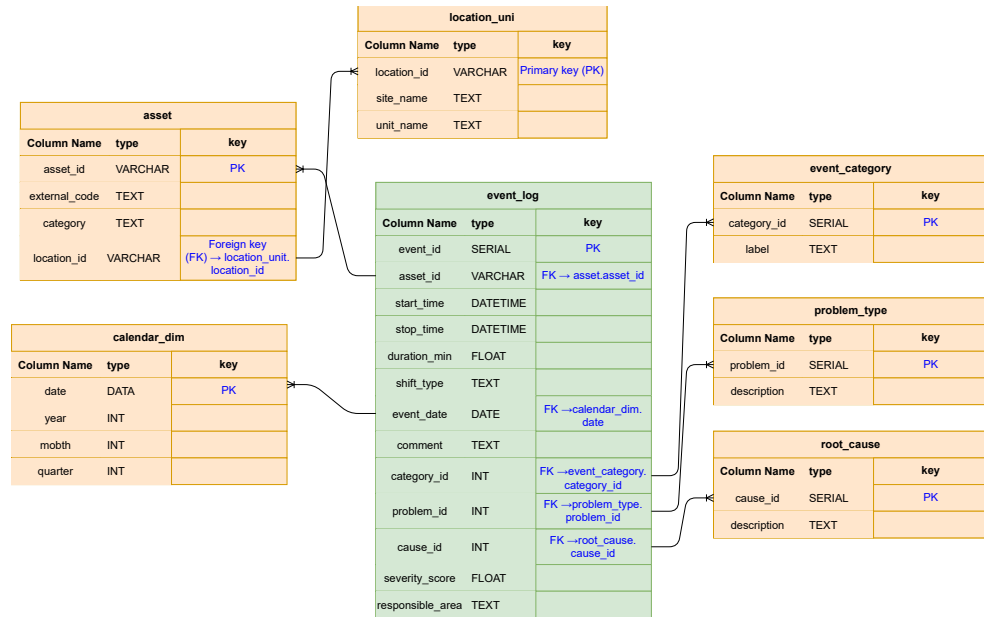


Fig. B5: Normalized relational schema for logging machine faults and operational events as the foundation for predictive maintenance.

Declarations

Ethics declaration

Not applicable.

Consent to Publish

Not applicable.

Consent to Participate

Not applicable.

Data Availability

The datasets generated and/or analysed during the current study are not publicly available due to confidentiality agreements with industrial project partners, but are available from the corresponding author on reasonable request.

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